

# Recipe determinants of quantization fragility under image corruption: paired evidence and a regression-head level-deficit localization in object detectors

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## ABSTRACT

Whether an 8-bit quantized detector stays reliable under corruption cannot be read off the format label: it depends on how the engine was built and which model region is quantized. We pair every corrupted input with a codec-matched clean control, so identical bytes reach executable TensorRT INT8 and FP8 engines, and estimate the corruption-associated change in the FP8–INT8 average-precision (AP) gap,  $\Delta E$ , inside common-image bootstrap samples. Across nine dataset×scale blocks of YOLO11 on KITTI, VOC, and a 2,000-image COCO subset,  $\Delta E$  is strongly heterogeneous ( $I^2 = 87\%$ ) and both signs survive Holm correction, even though FP8 exceeds INT8 on absolute clean and corrupted AP in every matched block mean. In RetinaNet, a matched-recipe decomposition localizes a severe INT8 level deficit (−21.2 AP on KITTI) to the regression head: leaving only that head in floating point restores clean-competitive accuracy without retraining (49.60 vs. 49.48 AP for matched FP8 on original clean inputs) and matches FP8 on the KITTI corrupt mean, leaving a residual that widens under corruption to about one AP point on VOC. A two-fold intervention, each fold frozen before its own runs, then calibrates INT8 on corrupted images and tests unseen families. The frozen held-out criterion—read directionally, so a significant harm counts as failure—is met in two of four fold×dataset cells, with a significant across-the-board harm in a third; a corruption-specific held-out benefit is significantly positive in exactly one cell, on both intervention tiers. Codec-matched calibration controls (post hoc for fold 1, pre-specified for fold 2) bound the codec level contribution far below the INT8 level deficit, though codec interaction effects are the same order as the corruption-specific components. A post-hoc operand-level factorial inside the head attributes essentially all of the deficit to activation quantization—INT8 head weights contribute at most a  $\sim 0.2$ -AP residual—with an operand interaction compatible with zero on the clean basis, while the same protocol on anchor-free FCOS shows no head-specific deficit—head exclusion changes nothing measurable—showing the fragility is not a generic property of INT8 detectors. Three conclusions follow: regression-head exclusion (or, in the tested recipe, promoting only the head’s activations) repairs the observed level deficit in the tested recipe without retraining (deployment cost not evaluated, and its strongest check is development-contaminated); corruption-aware calibration is inconsistent for held-out transfer; and the mechanism is activation-side but the operative failure mode—range clipping versus bin resolution—remains untested. Quantization robustness should be reported per recipe and per model region.

## 1. Introduction

Object detectors deployed on edge accelerators increasingly rely on post-training quantization (PTQ) to meet memory, energy, and latency budgets [1–5]. Two 8-bit formats dominate this setting: INT8, with a mature integer-arithmetic inference path [4, 6, 7], and FP8, which trades mantissa precision for exponent range on modern hardware [8, 9]. At the same time, deployment images deviate from benchmark conditions through noise, blur, weather, and compression artifacts that measurably degrade detector accuracy [10–13]. A natural question follows: which quantized format tolerates corruption better? We argue that this question, as usually asked, is ill-posed, because “INT8” and “FP8” name datatypes, whereas an executable detector is determined by a *recipe*: calibration data and preprocessing, quantizer placement and operator

coverage, excluded regions, graph transformations, and runtime settings.

Two confounds make naive format comparisons misleading. First, a gap measured only on corrupted images mixes the accuracy difference already present on clean images with the change induced by corruption; when one format also wins on clean images, the corrupted-input gap overstates the corruption effect. Second, recipes can differ systematically across formats in published pipelines: INT8 engines may be built with calibration distributions, preprocessing contracts, and operator-exclusion rules that differ from their FP8 counterparts—our own historical engines are one such instance (Section 4.2), and the two formats’ different numerical structure invites different recipe choices [14]—so a “format effect” may in fact be a recipe effect. A credible comparison must therefore pair treatments

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on identical image bytes, subtract the matched clean difference, and—crucially—vary the recipe deliberately rather than only the datatype.

This paper provides such a study on executable TensorRT engines evaluated end-to-end, with the decoder, non-maximum suppression (NMS), and evaluator held fixed within each detector family. We define a paired estimand  $\Delta E$ : the change in the FP8–INT8 AP gap between a JPEG quality 95 clean control and each corruption cell, estimated inside common-image bootstrap resamples so that pairing covariance is preserved. Under a frozen analysis contract—the estimands, condition schedule, and bootstrap plan are versioned before the paired analysis, while the YOLO11 engines themselves predate it and are disclosed as such (Section 3.4)—covering YOLO11n/m/x [15] on KITTI, VOC, and a deterministic 2,000-image COCO subset with 14 conditions per block, we find that  $\Delta E$  reverses sign across dataset×model combinations (Section 4.1). The format label alone does not determine corruption behavior.

The sharpest evidence comes from a second detector family. On RetinaNet (ResNet-50 feature-pyramid-network (FPN) backbone) [16, 17], the historical INT8 recipe collapses on KITTI (−42.3 AP relative to a matched FP8 engine); repairing its calibration preprocessing contract still leaves a −21.2-AP deficit for matched INT8, and a stage-wise recipe decomposition traces that residual to one model region: quantizing only the classification head is harmless, quantizing only the regression head reproduces the collapse, and excluding the regression head alone restores clean-competitive INT8 accuracy (49.60 AP, matching FP8-matched at 49.48) without retraining (Section 4.2). The matched collapse is much milder on VOC (−2.9 AP), so the fragile region’s contribution is itself dataset-dependent.

Finally, we turn the diagnostic into a controlled two-fold intervention. If corrupted-input accuracy were limited by calibration on clean images only—ranges that fail to cover corrupted-input activations—then calibrating on corruption-augmented images should restore accuracy on held-out corruption types. Freezing complementary splits (Gaussian noise and JPEG calibrated with fog and motion blur held out, and vice versa) shows that the intervention itself is context-dependent: the frozen held-out criterion is met on KITTI fold 1 (+1.07 AP) and VOC fold 2 but fails on the other two cells, including a significant across-the-board harm on KITTI fold 2, and a corruption-specific held-out component is significantly positive in exactly one cell. Three conclusions follow (Section 4.4): excluding the regression head is a validated retraining-free repair of the observed level deficit in the tested recipe; corruption-aware calibration is context-dependent and is not supported as a reliable repair of corrupted-input accuracy; and neither intervention isolates the mechanism behind the head-level failure, which remains open.

This study makes five contributions:

- A paired, codec-controlled evaluation protocol for executable quantized detectors whose estimand isolates the corruption-associated change in a between-treatment

gap, with pairing preserved by common-image bootstrap samples.

- Evidence, with a formal heterogeneity test and multiplicity control, that the corruption interaction is context-dependent within one detector family:  $\Delta E$  changes sign across dataset and model-scale blocks, so the corruption-induced change in the gap has no stable sign on this protocol, while FP8 exceeds INT8 on absolute AP in every matched block mean.
- A recipe decomposition that localizes the RetinaNet INT8 collapse to regression-head quantization, sharpened by a post-hoc operand factorial to the head’s *activation* tensors (INT8 head weights contribute at most a  $\sim 0.2$ -AP residual; the weight×activation interaction is compatible with zero), with a non-replication on anchor-free FCOS showing the fragility is not general to INT8 detectors.
- A retraining-free selective-precision INT8 recipe—only the regression head kept in floating point—whose exclusion direction is validated under the frozen analysis contract on both datasets, while the >20-AP magnitude is a KITTI estimate carrying a flagged development-contamination caveat and a significant VOC residual is reported (deployment cost unmeasured).
- A controlled corruption-aware calibration intervention over two complementary in-family/held-out folds with codec-matched controls, analyzed with the same paired estimand: the intervention’s held-out effect is itself sign-unstable across fold×dataset cells, so calibration coverage contributes context-dependently and is not supported as a reliable repair of corrupted-input accuracy.

Section 2 reviews quantization under corruption, detector-specific PTQ, and evaluation methodology. Section 3 defines the protocol, treatments, datasets, and the frozen intervention design. Section 4 reports the interaction landscape, the RetinaNet decomposition and its operand-level factorial, the FCOS second-architecture check, and the intervention outcome. Section 5 discusses mechanism, practical guidance, and limitations. Section 6 concludes.

## 2. Related work

### 2.1. Post-training quantization for detectors

PTQ estimates numerical ranges with representative data or data-free surrogates and reduces quantization error through clipping, rounding, or local reconstruction [1, 18–23]. Detector-specific PTQ is harder than classification because classification confidence, box localization, feature pyramids, and foreground–background imbalance [24] must remain jointly calibrated: PD-Quant uses prediction-difference losses, DetPTQ a detection-output loss, and Reg-PTQ specializes calibration and quantization for regression structures [25–27]. InlierQ suppresses task-irrelevant activation anomalies using a small unlabeled calibration set [28]. These methods optimize clean-input accuracy; our decomposition instead asks which structural component

of a deployed INT8 detector is responsible for fragility, a question Reg-PTQ’s emphasis on regression structures anticipates but does not measure at the executable-engine level.

Quantization-aware training and distillation approaches [29–32] can produce robust low-precision detectors but require retraining. Our selective-precision arm is deliberately retraining-free: it changes which regions of the graph are quantized, keeping the deployed checkpoint and calibration budget fixed, which makes the recipe contrast interpretable.

## 2.2. INT8, FP8, and executable treatments

Integer-only inference is an established path [4, 6, 33], while FP8 formats evolved from training formats [34, 35] to practical PTQ layouts [8, 9, 36, 37]. Comparisons show that accuracy and hardware efficiency can favor different representations and that the preferred 8-bit format varies by layer and architecture [14, 38]. The nominal datatype therefore does not describe an executable engine: quantizer coverage, scale resolution, excluded operators, calibration, and runtime all contribute. Our protocol records the complete ModelOpt/TensorRT treatment and manipulates recipe components one at a time, which is what allows the recipe-format confound to be measured rather than assumed.

## 2.3. Corruption robustness and quantized models

Corruption benchmarks [10] measure aggregate degradation for classifiers, and detector studies report task-specific failure under weather, noise, blur, and compression [11–13]. The standard summary normalizes mean performance under corruption (mPC) by clean performance ( $P_{\text{clean}}$ ) as the relative rate  $\text{rPC} = \text{mPC}/P_{\text{clean}}$  [11], which answers a within-model retention question. Our estimand instead asks how the *between-treatment* AP-point gap changes on identical corrupted bytes; a related absolute-AP guard prevents a shrinking gap near a shared failure floor from being read as robustness.

Work linking quantization and distribution shift is comparatively thin: Karimov et al. evaluate quantized detectors under degraded inputs and report sensitivity to the evaluation setting [39]; a separate line of work measures quantization’s effect on *adversarial* robustness [40], a distinct shift family. TTAQ adapts quantization during continual shift [41], Recti-Q corrects out-of-distribution quantized features [42], and test-time adaptation more broadly updates the model at inference, including normalization-statistics adaptation under corruption [43–46]. Corruption-aware training and augmentation improve robustness by exposing optimization to degraded inputs [47, 48], and detector error analyses show that conclusions depend on how errors are decomposed [49, 50]; evaluation-protocol work likewise shows that evaluation design itself changes system-level conclusions [51, 52]. Our intervention is related in spirit to corruption-aware calibration, but differs in the experimental control: the calibration corruption families are disjoint from the held-out evaluation families, so any claimed robustness benefit must generalize rather than memorize.

## 3. Methodology

### 3.1. Executable treatments as recipes

Let a *treatment* denote the complete recorded configuration of an executable detector: checkpoint, Open Neural Network Exchange (ONNX) graph, quantizer placement, quantize/dequantize (Q/DQ) coverage, calibration list, calibration preprocessing, excluded operators, TensorRT builder settings, runtime, and decoder. Throughout, “INT8” and “FP8” qualify such treatments, not isolated datatypes, and an *arm* denotes one treatment instance within an experiment. All engines were built with TensorRT 11.1.0.106 [53] on an NVIDIA GeForce RTX 5090 with a 4,096 MiB builder workspace, from ONNX graphs quantized by NVIDIA ModelOpt using entropy calibration [54, 55]; the study-built registries record ModelOpt 0.45.0, while the pre-study YOLO11 ONNX registries predate that version field.

Builder policy differs across the two evidence programs and is itself a recipe attribute, recorded per engine in the manifests: the YOLO11 engines predate the frozen contract and were built by `trtexec` with TensorRT defaults (TF32 enabled), while the RetinaNet quantization arms built for this study use the TensorRT Python builder with TF32 disabled (`tf32: false` in each engine registry). The RetinaNet FP32 reference and the FP8/INT8-legacy arms are pre-study `trtexec` builds that carry TensorRT defaults, so TF32 is enabled there; within each family the TF32 state is identical across the paired *quantized* arms, which are the only arms entering the  $\Delta E$  contrasts. Calibration lists contain 512 deterministically selected training images for all paired and intervention arms (the earlier diagnostic sweeps include 128-image and max-estimator variants, marked in the diagnostic table of Section 4.2); evaluation and calibration images are disjoint.

Within this framing we define the RetinaNet recipe arms used below:

- **INT8-legacy**: the historical transferred recipe; its calibration preprocessing does not match inference preprocessing (a contract violation retained as a diagnostic arm).
- **INT8-matched**: entropy INT8 with calibration preprocessing identical to inference (`retinanet_normalized`) and the same 512-image budget as FP8.
- **INT8-selective**: INT8-matched with all `/head/regression_head/` nodes left in floating point—a selective-precision recipe requiring no retraining.
- **INT8-corruptcalib** and **INT8-sel-corruptcalib**: the matched and selective recipes rebuilt on a corruption-aware calibration set (noise and JPEG; Section 3.6); **INT8-cc2calib** and **INT8-sel-cc2calib** are the complementary fold (fog and motion blur).
- **INT8-q95calib** and **INT8-sel-q95calib**: codec-matched calibration controls—the same 512 clean images re-encoded at JPEG quality 95 with no corruption (Section 3.6).



**Figure 1:** Paired evaluation protocol. A deterministic clean control encoded at JPEG quality 95 (J95) and each corrupted input are created before the precision branch. The four paired cells combine INT8 and FP8 treatments with J95-clean and corrupted inputs; the FP32 engine is a separate reference and cancels from the interaction  $\Delta E$ . Common image resamples preserve pairing. Clean-control fidelity and absolute corrupted AP—and the signed loss  $D = AP_{J95} - AP_c$  shown in the floor guard—are checked before an interaction is interpreted.

- **FP8-matched** and **FP32:** matched-recipe FP8 and the high-precision reference.

One further pre-study engine is cited as context only and is outside this arm list: the **FP8-legacy** exploratory build (a `trtexec-default`, TF32-on engine evaluated under the frozen protocol for reference). INT8-legacy above is likewise pre-study but enters the frozen protocol’s arm table as the documented contract-violation arm. Three senses of “matched” recur and are kept distinct: *contract-matched* arms use calibration preprocessing identical to inference; *codec-matched* inputs share the terminal JPEG quality-95 encoding; and *paired* comparisons evaluate treatments on identical images (and identical resamples in the bootstrap).

### 3.2. Paired estimands and interpretation order

For dataset  $d$ , model  $m$ , quantized treatment  $q$ , and condition  $c$ , write  $AP_{d,m,q,c}$  for COCO-style  $AP@[.5:.95]$  reported in AP points. J95 denotes the clean control: the source image deterministically encoded at JPEG quality 95 with subsampling disabled and no named corruption (Fig. 1). The between-treatment gaps on clean and corrupted inputs are

$$G_{d,m,x} = AP_{d,m,FP8,x} - AP_{d,m,INT8,x}, \quad x \in \{J95, c\}, \quad (1)$$

and the primary interaction is

$$\Delta E_{d,m,c} = G_{d,m,c} - G_{d,m,J95}. \quad (2)$$

Positive  $\Delta E$  means the FP8–INT8 gap widens under corruption relative to the clean baseline; negative means it narrows. The FP32 reference cancels algebraically. We abbreviate

$G_0 = G_{d,m,J95}$  and write  $G_{\text{corr}}$  (“corrupt-mean gap”) for the equal-weighted mean of  $G_{d,m,c}$  over the 12 corruption-severity cells; the reported per-block  $\Delta E$  is  $G_{\text{corr}} - G_0$  (Eq. (2)). Absolute corrupted AP is reported beside  $\Delta E$ ; a shrinking gap near a shared accuracy floor is a floor effect, not robustness (the “floor guard”: interpretations are made jointly with the corrupted AP level). The contrast is descriptive, not causal—algebra analogous to a difference-in-differences interaction [56] without causal identification claims.

The same construction applies to recipe contrasts: substituting two arms of the same format (e.g., INT8-selective minus INT8-matched) isolates a recipe component under an identical protocol, and restricting the corrupted set to a subset  $S$  gives the analogue used for the intervention contrasts:

$$\Delta E_{A-B,S} = [AP_A - AP_B]_S - [AP_A - AP_B]_{J95}, \quad (3)$$

written  $\Delta E_{\text{held-out}}$  and  $\Delta E_{\text{in-family}}$  for the intervention’s two partitions. The sign is read with the contrast order: for an intervention-minus-baseline contrast, *positive* is a corruption-specific benefit to the intervention arm; under the FP8–INT8 order of Eq. (2), positive instead means the FP8 advantage grows. The sign convention is restated beside each use. This substitution is the key methodological step that turns a format comparison into a recipe experiment.

Two quantities are kept distinct throughout: an arm’s *level deficit*—its AP shortfall to a comparator on a given input set, present already on clean inputs for the RetinaNet

INT8 arms (its magnitude equals  $G_0$  when the comparator is the paired FP8 arm)—and the corruption-specific interaction  $\Delta E$ . The RetinaNet INT8-matched arm exhibits a large level deficit but a negative  $\Delta E$  on KITTI, so Sections 4–5 keep the two distinct: “level deficit” and “collapse” for the AP shortfall an arm already carries, and “corruption-specific” language ( $\Delta E$ , interaction) for what corruption adds on top; conflating them would judge the intervention against a target it was never designed to hit.

### 3.3. Corruption suite and codec control

Four synthetic corruptions are generated deterministically: Gaussian noise ( $\sigma \in \{8, 20, 40\}$  on the 0–255 scale), linear motion blur (kernel lengths  $\{5, 13, 23\}$  at a per-image uniform angle), synthetic fog (strength–radius pairs  $(0.18, 6)$ ,  $(0.38, 12)$ ,  $(0.62, 20)$  blended over a per-image random field), and JPEG compression (qualities  $\{70, 40, 15\}$ ). Severity labels 1/3/5 order settings within each type. Every output is re-encoded at JPEG quality 95, so the clean control and corrupted inputs share the terminal codec; a SHA-256-derived seed binds dataset, image identifier, corruption, and severity. Within each detector family, inference fixes the input resolution (640-pixel letterbox), confidence threshold, decoder, and non-maximum suppression (NMS): YOLO11 uses a confidence threshold of 0.001 and class-wise NMS at intersection-over-union (IoU) 0.7 with at most 300 detections; RetinaNet uses 0.05 and batched NMS at IoU 0.5, also capped at 300 detections. AP evaluation uses the COCO protocol (pycocotools) with `maxDets=100`. Decoder constants therefore differ across families, and cross-family comparisons compare recipe bundles rather than a shared decoder; all within-family contrasts share identical decoder settings.

### 3.4. Datasets and the frozen paired protocol

Three evaluation universes anchor the frozen paired analysis: KITTI validation (1,496 images) [57], PASCAL VOC2007 test (4,952 images) [58], and a deterministic 2,000-image subset of COCO val2017 [59] (seed 20260924; the subset is identical across all arms so that COCO comparisons remain paired). VOC and KITTI annotations were converted once to COCO format and frozen; reported values are COCO-style measurements on converted annotations. One checkpoint per dataset×model cell is used throughout, and its provenance is disclosed rather than idealized: the KITTI and VOC YOLO11 scales were fine-tuned from Ultralytics pretrained weights under the pilot program (runs `*_train_v1`), with the best checkpoint selected on the same validation partition that serves as the evaluation universe here—so KITTI/VOC absolute AP levels carry selection optimism whose residual effect on paired contrasts is unquantified; COCO YOLO11 uses the official pretrained checkpoint with no dataset fine-tune; the RetinaNet ResNet-50-FPN-v2 model was fine-tuned per dataset for 100 epochs at 640-pixel input (seed 20260807) with best-checkpoint selection by minimum validation loss on the same evaluation partition, under the same caveat. All checkpoint, split, and training-run manifests are hash-bound

(Supplement S4). Each dataset×model block evaluates clean, codec-control, and the 12 corruption–severity cells on identical image ordering: 14 conditions per arm, except the RetinaNet FP32 and legacy arms, which predate codec control and carry 13. RetinaNet arms add the intervention engines on the same protocol (Section 3.6).

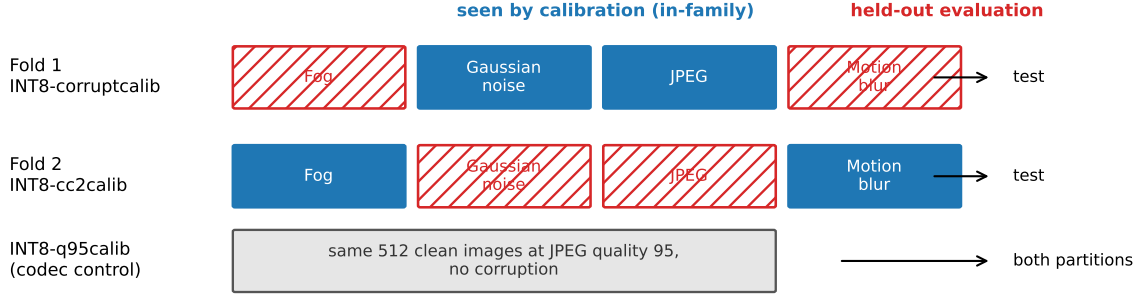
Two scope notes qualify interpretation. First, the KITTI validation universe was used while developing the selective-precision recipe, so KITTI selective-arm numbers are development-contaminated; VOC is the cleaner cross-dataset check for that arm. Second, coverage is asymmetric across families: RetinaNet runs on KITTI and VOC, YOLO11 on all three universes, so COCO evidence is single-family and cross-family comparisons on KITTI/VOC share the same checkpoint-selection caveat.

### 3.5. Paired image bootstrap

Uncertainty comes from the finite evaluation images with checkpoints, engines, calibration, decoder, and corruptions fixed; evaluation-set composition alone is a known source of score variance [60, 61]. Within each dataset×model block we use one deterministic schedule of 2,000 bootstrap resamples of image indices [62] shared across all arms and all conditions, so every paired contrast is evaluated inside common-image samples and the induced covariance is preserved. Reported point estimates are full-sample plug-in values; reported intervals are 2.5th–97.5th percentiles of the bootstrap distribution of each contrast, marked \* when they exclude zero. Two-sided bootstrap  $p$ -values,  $p = \min\{1, 2 \min[\Pr(d \leq 0), \Pr(d \geq 0)]\}$  (smallest nonzero value 1/1000 at  $B = 2000$ ), are Holm-adjusted across the nine paired YOLO11 blocks [63]. Heterogeneity of  $\Delta E$  across blocks is tested with Cochran’s  $Q$  on inverse-variance weights from the bootstrap standard errors [64, 65]; because blocks within a dataset share images, their estimates may be correlated; the cross-block covariance is not estimated, so  $Q$  is an independence approximation whose error direction is not established (shared-image resampling plausibly induces positive covariance, which would make  $Q$  conservative, but this is not verified). Diagnostic decomposition arms are single-point estimates without intervals; the exploratory layer’s cells carry their own bootstrap intervals in the retained ledger and are reported only as context (Section 4.5).

### 3.6. Corruption-aware calibration intervention

The intervention tests one hypothesis under decision rules frozen in a versioned configuration before any intervention run: *if the INT8-matched arm’s corrupted-input accuracy is limited by calibration images that never exhibit corrupted-input activation ranges, then corrupting the calibration distribution should recover held-out corruption AP*. The hypothesis is deliberately scoped to corrupted-input accuracy: it cannot by itself account for the clean-image level deficit, which the decomposition attributes elsewhere (Section 4.2). Each of the 512 calibration images is deterministically assigned—by a hash of dataset and image path under schedule `nn-corruptcalib-v1`—to exactly one of



**Figure 2:** Corruption-calibration folds and the codec-matched control arm. Filled blue cells are families seen by entropy calibration (the in-family partition); hatched red cells are held out and never calibrated. Fold 1 (INT8-corruptcalib) calibrates Gaussian noise and JPEG and is tested on fog and motion blur; fold 2 (INT8-cc2calib) swaps the roles. INT8-q95calib sees only the same 512 clean images re-encoded at JPEG quality 95 and is evaluated on both partitions. Selective-tier arms (sel-corruptcalib, sel-cc2calib, sel-q95calib) share each fold’s split.

**Table 1**

Intervention decision rules and their registration status. “dir.”: the directional reading reported in the text—a significant harm counts as failure. “null-seek”: met only when the interval contains zero (a bound, not an equivalence proof). <sup>a</sup>the registered wording is two-sided (CI excluding 0, either direction); the text reports the directional reading. <sup>b</sup>recorded as a frozen primary estimand, not a success criterion. Codec-matched (q95calib) controls are post hoc for fold 1 and pre-specified anchors in fold 2.

| Contrast                           | Tier | Fold 1                       | Fold 2                    |
|------------------------------------|------|------------------------------|---------------------------|
| Held-out gain vs clean-calib. arm  | full | frozen, dir.                 | frozen, dir. <sup>a</sup> |
| Held-out $\Delta E$ vs matched/q95 | full | post hoc                     | frozen, dir.              |
| Held-out gain vs clean-calib. arm  | sel. | frozen estimand <sup>b</sup> | post hoc                  |
| Held-out residual vs FP8 incl. 0   | sel. | frozen, null-seek            | post hoc                  |

six in-family cells (77–102 images per cell) and corrupted with the same frozen parameters and JPEG quality-95 output encoding as the evaluation caches (Section 3.3). Figure 2 shows the two complementary splits and the codec-matched control: fold 1 calibrates Gaussian noise and JPEG and holds out fog and motion blur, which are never calibrated; fold 2 swaps the roles on the same 512 images (75–95 images per cell). The full tier re-quantizes INT8-matched; the selective tier applies the same calibration to INT8-selective.

Table 1 records the decision rules and when each was registered. The fold-1 configuration carries two success criteria: (i) the full-tier corrupt-calibration arm improves the *held-out* corrupt mean over INT8-matched with an interval excluding zero on the positive side, which would support the range-mismatch hypothesis; and (ii) the selective corrupt-calibration arm’s *held-out* mean *includes* zero relative to FP8-matched, which would let calibration explain the VOC residual gap of Section 4.2—deliberately null-seeking, since no equivalence margin was pre-specified. The fold-1 primary estimand additionally covers the selective-tier *held-out* gain over its clean-calibrated counterpart. The complementary fold 2 is frozen in its own configuration before any of its runs and pre-specifies for the full tier (INT8-cc2calib)

a replication test of the fold-1 *held-out* criterion and the corruption-specific rule— $\Delta E_{\text{held-out}}$  of Eq. (3) against both the INT8-matched and the INT8-q95calib anchors, excluding zero on the positive side—while its selective-tier contrasts are post hoc.

One wording subtlety matters for how fold-2 outcomes are scored. The fold-2 primary rule is worded non-directionally in the configuration (“CI95 excluding 0  $\Rightarrow$  fold-1 frozen criterion replicated”): the literal registered test detects a *held-out* change in either direction, and we read “replicated” through the directional fold-1 criterion it cites, so a significantly *negative* *held-out* change counts as failure. Under the literal two-sided wording, both fold-2 cells would register as “replicated”; the directional reading changes only the KITTI verdict. The directional reading is used throughout; the verdict paragraph reports the literal count as well. In-family improvement alone is evidence of calibration-fitting, not robustness transfer, and all rules are reported regardless of outcome.

Two interpretive analyses accompany the frozen rules and are labeled as such because they were added after the fold-1 freeze (fold 2 subsequently froze the  $\Delta E$  rule, but not the magnitude comparison): (i) a magnitude comparison of any *held-out* gain against the residual level deficit to matched FP8, and (ii) the corruption-specific decomposition  $\Delta E_{\text{held-out}}$  of Eq. (3), which separates a corruption-specific benefit from a uniform accuracy shift that would appear on clean inputs too. This subtraction rests on an additive, parallel-shift assumption—that any calibration-induced level shift on clean inputs transfers comparably to corrupted inputs—so it is an interpretive decomposition, not a causal identification (Supplement S7).

Two controls discipline interpretation. First, the corrupt-calibration images carry the same JPEG quality 95 terminal encoding as the evaluation caches, whereas the matched clean-calibration list points at original bytes; a gain of the intervention arm over INT8-matched is therefore potentially confounded by codec adaptation rather than corruption coverage. We isolate this component with the INT8-q95calib

and INT8-sel-q95calib control arms, calibrated on the same 512 images re-encoded at JPEG quality 95 without corruption. These codec controls were added after the fold-1 freeze and after the fold-1 results were computed, when the confound was identified during review of the intervention design; they are therefore post-hoc controls on the fold-1 contrast and are labeled as such, while fold 2 pre-specifies its  $\Delta E$  rule against the q95-matched anchor in advance. Second, the codec confound works in the direction that matters for the primary test: a confound that *inflates* the intervention arm’s gains biases the primary superiority test *toward* the positive verdict, so a positive held-out result cannot be attributed to corruption coverage until it survives against the codec-matched anchor. For the null-seeking selective criterion the direction of the confound reverses: codec adaptation that lifts the selective arm moves its negative held-out residual toward zero, so it could in principle manufacture an interval containing zero—a met criterion would be suspect, while an observed failure is conservative.

### 3.7. Post-hoc extensions: operand-level decomposition and a second-architecture replication

Two further arms strengthen the localization of Section 4.2 without retraining. *First, an operand-level factorial*: starting from the frozen INT8-matched QDQ graph, we derive two rewired variants of the regression head in which only one operand class is quantized—INT8 weights with fp16-compute activations (W-only) and fp16 weights with INT8 activations (A-only)—by rewiring the head’s quantize/dequantize edges in the ONNX graph while leaving every other node, weight initializer, and calibration output byte-identical to the matched arm. Together with the matched (INT8 weights, INT8 activations) and selective (both operands fp16) arms this gives a 2×2 factorial over head weights × head activations, so the clean-level deficit can be attributed to weight quantization, activation quantization, or their interaction rather than to the head as a whole. Each rewired graph is validated by diffing every rewired edge against the corresponding edge treatment in the pre-existing selective-arm graph (fp16 initializer for bypassed weights; direct tensor bypass for dequantized activations), so the arms differ from matched only in which operand the head consumes in quantized form. Rewiring is consumer-local, so “head activations” here means every activation tensor *consumed by* a regression-head node—including the FPN feature maps at the head’s input boundary—not only tensors produced inside the tower; the factorial therefore attributes the deficit to activations in or at the head rather than to internal head activations alone. *Second, a replication on a different detector family*: we train anchor-free FCOS (ResNet-50-FPN) on the same KITTI and VOC partitions under the same training profile, export the raw head outputs to ONNX, and run the same matched/selective arm set—the selective exclusion covers FCOS’s regression tower including its centerness output, since torchvision places `bbox_ctrness` inside

**Table 2**

Evidence layers and their inferential status. Intervals exist only where pairing and the frozen schedule are defined.

| Layer                    | Evidence                           | Engines/protocol                                             | Uncertainty                        |
|--------------------------|------------------------------------|--------------------------------------------------------------|------------------------------------|
| Diagnostic               | Region-wise recipe sweeps          | Development builds                                           | Single-point AP                    |
| Frozen paired analysis   | Paired 14-cell protocol            | Pre-existing engines under a frozen analysis contract        | Paired bootstrap (2,000 resamples) |
| Intervention             | Calibration interventions          | Frozen fold config; fold-1 codec controls post hoc           | Paired bootstrap (2,000 resamples) |
| Historical exploratory   | 144-cell prior grid                | Shared engines; separate retrained-holdout and legacy builds | Reported as context                |
| Operand factorial + FCOS | Post-hoc arms on frozen conditions | QDQ-revised and newly trained engines, unregistered          | Paired bootstrap (2,000 resamples) |

head.regression\_head. The FCOS decoder was validated detection-for-detection against torchvision’s reference post-processing on 32 seeded synthetic head-output draws—identical detection sets, zero maximum box and score deviation (Supplement, `decoder_validation.json`; a post-processing math check, not an end-to-end engine parity test); the RetinaNet and FCOS checkpoints (one per detector×dataset) are all early minimum-validation-loss picks (Supplement); the VOC FCOS training loss diverged to NaN at epoch 70, long after its epoch-10 selection point. Both additions are post hoc relative to the paired-protocol freeze: they evaluate frozen conditions under the same paired bootstrap schedule but the arms themselves were fixed after the paired freeze, so we treat them as post-hoc extensions and report them regardless of outcome.

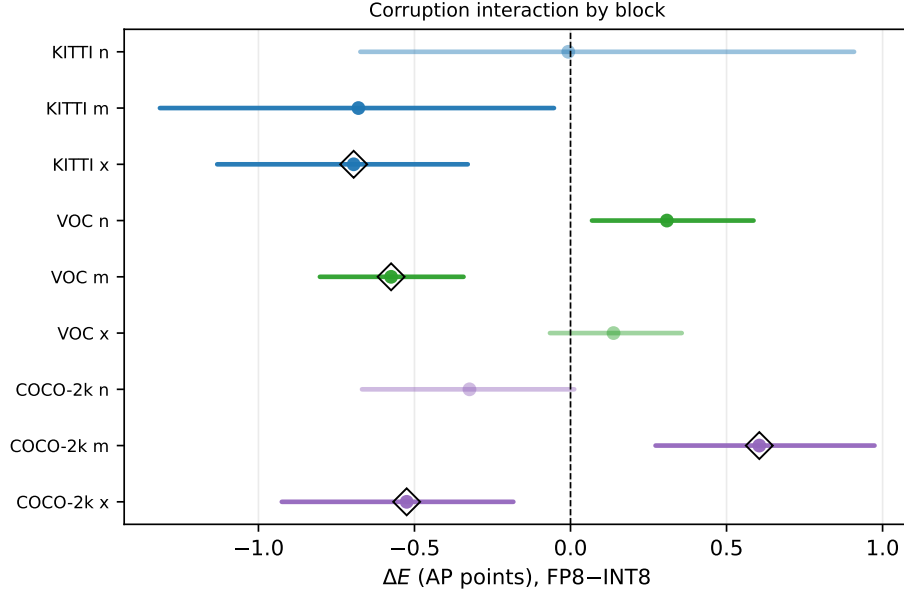
### 3.8. Evidence layers

Five evidence layers are kept distinct (Table 2): *diagnostic* evidence (Section 4.2) consists of single-point recipe sweeps that localized the fragile region during development; *frozen paired analysis* is the paired protocol of Section 3.4 with bootstrap intervals—the freeze applies to the evaluation and analysis contract (estimands, condition schedule, bootstrap schedule), not to the artifacts, since the YOLO11 engines, predictions, and evaluation partitions were built for the exploratory program and are shared with the historical layer; *intervention* evidence is the corrupt-calibration experiment and its codec controls (Section 3.6), whose two corruption-calibration folds were frozen prospectively before their engines were built while the fold-1 codec-control arms are post-hoc additions. A fourth, historical exploratory layer—the same paired YOLO11 engines and estimand analyzed earlier as a 144-cell grid over four datasets with a different scope and analysis plan—is reported only as context (Section 4.5). A fifth layer—the operand-factorial and FCOS arms of Section 3.7—runs the frozen paired conditions on post-hoc arms and is reported separately from the frozen content.

## 4. Results

### 4.1. The FP8–INT8 interaction is not universal

Table 3 and Fig. 3 report the paired YOLO11 landscape: nine dataset×model blocks, each with the clean-control gap  $G_0$ , the corrupt-mean gap  $G_{\text{corr}}$ , and the paired interaction  $\Delta E$  with its 95% interval. FP8 exceeds INT8 on clean-control AP in all nine blocks ( $G_0$  from +0.68 to +3.97 AP points) and on the 12-cell corrupt mean as well (FP8 and INT8 corrupted AP columns, Table 3); the gap is positive on the level in every block, although a



**Figure 3:** Paired corruption interaction  $\Delta E$  (FP8–INT8) with 95% common-image-bootstrap intervals. Solid markers denote intervals excluding zero (unadjusted); open diamonds mark blocks surviving Holm adjustment across the nine blocks ( $p_{\text{Holm}} < 0.05$ ). The sign reverses across dataset×model blocks, so the interaction cannot be summarized as a single format-level effect.

minority of individual severity cells reverse it. What lacks a stable sign is the change corruption induces in that gap: it is positive on VOC/YOLO11n (+0.31 [+0.07, +0.59]) and COCO-2k/YOLO11m (+0.60 [+0.27, +0.97]), negative on VOC/YOLO11m (−0.57 [−0.80, −0.34]), KITTI/YOLO11m (−0.68 [−1.32, −0.05]), KITTI/YOLO11x (−0.69 [−1.13, −0.33]), and COCO-2k/YOLO11x (−0.53 [−0.93, −0.18]), and compatible with zero on the remaining three blocks (unadjusted intervals; multiplicity is addressed below). Within VOC alone, the sign of the estimate flips with model scale (n : +0.31 excluding zero; m : −0.57 excluding zero; x : +0.14 null). A claim that INT8 is uniformly less robust under corruption—or uniformly comparable—is not supported on this protocol’s own evidence; within a fixed recipe, the sign of the interaction varies with the dataset×model context.

The heterogeneity is formal, not an artifact of reading nine intervals. Cochran’s test rejects a common  $\Delta E$  across blocks ( $Q = 63.3$  on 8 degrees of freedom,  $p \approx 1 \times 10^{-10}$ ;  $I^2 = 87\%$ , between-block standard deviation  $\tau = 0.44$  AP), so a pooled “format effect” would obscure substantial context dependence. The heterogeneity is not only between datasets: it holds within VOC ( $Q = 29.3$ , 2 df,  $p \approx 4 \times 10^{-7}$ ) and within the COCO subset ( $Q = 21.9$ , 2 df,  $p \approx 2 \times 10^{-5}$ ), where only model scale varies, though not within KITTI ( $Q = 2.4$ ,  $p = 0.30$ ). After Holm adjustment across the nine blocks, 4 blocks remain significant and they carry *both* signs—one positive (COCO-2k/YOLO11m) and three negative (KITTI/YOLO11x, VOC/YOLO11m, COCO-2k/YOLO11x; Table 3); a DerSimonian–Laird pooled mean is −0.18 AP [−0.50, +0.13], a Wald interval—a descriptive average of heterogeneous interactions, not

a common effect. Within one dataset, VOC/YOLO11n and VOC/YOLO11m differ by +0.88 AP ( $z = 4.9$  under an independence approximation that is conservative for positively correlated blocks). The heterogeneity also exists *within* blocks at the family level (Fig. 4)—on VOC/YOLO11x the JPEG family contributes −1.07 AP while Gaussian noise and motion blur contribute +0.69 and +0.67 AP—and excluding the saturated motion-blur-s5 cell leaves the cross-block heterogeneity essentially unchanged ( $Q \approx 60.2$ ,  $I^2 \approx 87\%$ ) and every Holm-surviving block’s sign intact, although the two KITTI blocks that do not survive Holm drift (YOLO11n’s null point estimate flips from −0.01 to +0.15 and YOLO11m’s unadjusted-significant negative turns null; Supplement S8, Table S2). The codec control also behaves as required: J95 clean AP tracks original clean AP within 0.3 points for every arm in these nine blocks, and the within-model retention statistic (corrupt mean over J95 clean) ranges 0.68–0.79 across blocks with INT8-versus-FP8 differences of at most  $\sim 0.02$ —the pairing basis itself does not penalize either treatment.

#### 4.2. A matched recipe exposes and localizes the RetinaNet level deficit

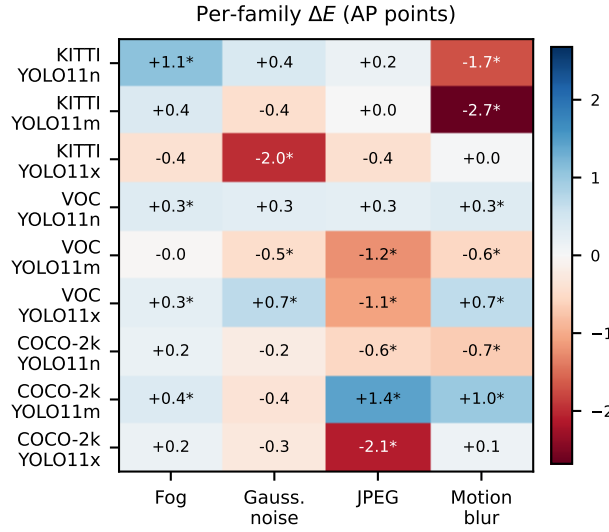
The non-YOLO comparator is where recipe effects dominate. Table 4 and Figs. 5–6 show the RetinaNet arms on KITTI and VOC under the frozen protocol. Five observations matter, plus a boundary condition.

First, about half of the historical INT8 recipe’s collapse is a contract violation, not a format property: in the controlled diagnostic—which rebuilds the mismatched contract under the study’s common no-TF32 builder policy at the

**Table 3**

Frozen paired-analysis contrasts, FP8 minus INT8, per dataset×model block. FP8/INT8 corr. AP: absolute 12-cell corrupt-mean AP (the floor guard);  $G_0$ : clean-control (J95) gap;  $G_{\text{corr}}$ : corrupt-mean gap (the difference of the two absolute columns, computed before they are rounded to 0.1 AP);  $\Delta E = G_{\text{corr}} - G_0$  with paired-bootstrap interval. \* interval excludes zero;  $p_{\text{Holm}}$ : bootstrap  $p$ -value Holm-adjusted across the nine blocks.

| Dataset | Model   | FP8 corr. AP | INT8 corr. AP | $G_0$ | $G_c$ | $\Delta E$ | 95% interval   | $p_{\text{Holm}}$ |
|---------|---------|--------------|---------------|-------|-------|------------|----------------|-------------------|
| KITTI   | YOLO11n | 44.1         | 41.9          | +2.14 | +2.13 | -0.01      | [-0.67, +0.91] | 0.914             |
| KITTI   | YOLO11m | 50.4         | 47.1          | +3.97 | +3.29 | -0.68*     | [-1.32, -0.05] | 0.124             |
| KITTI   | YOLO11x | 51.0         | 50.3          | +1.44 | +0.74 | -0.69*     | [-1.13, -0.33] | 0.008             |
| VOC     | YOLO11n | 44.8         | 43.8          | +0.68 | +0.99 | +0.31*     | [+0.07, +0.59] | 0.075             |
| VOC     | YOLO11m | 51.8         | 50.7          | +1.70 | +1.12 | -0.57*     | [-0.80, -0.34] | <0.009            |
| VOC     | YOLO11x | 54.3         | 53.3          | +0.91 | +1.05 | +0.14      | [-0.07, +0.36] | 0.354             |
| COCO-2k | YOLO11n | 28.2         | 27.4          | +1.10 | +0.78 | -0.32      | [-0.67, +0.01] | 0.168             |
| COCO-2k | YOLO11m | 38.0         | 36.5          | +0.89 | +1.49 | +0.60*     | [+0.27, +0.97] | 0.008             |
| COCO-2k | YOLO11x | 42.4         | 41.9          | +1.00 | +0.47 | -0.53*     | [-0.93, -0.18] | 0.018             |



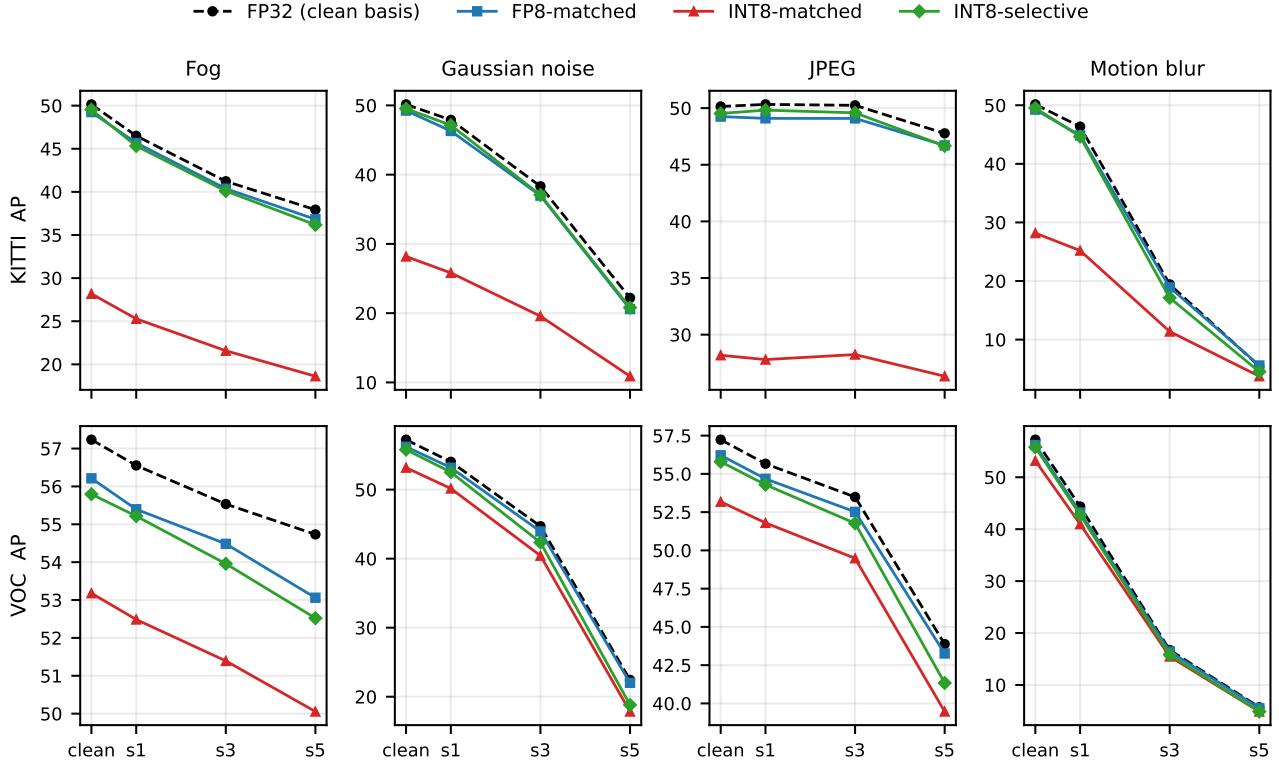
**Figure 4:** Per-family interaction  $\Delta E$  (FP8 minus INT8, on the J95-controlled basis) across the nine paired blocks, in AP points. Star: unadjusted 95% bootstrap interval excludes zero. Every corruption family carries both signs across blocks, and seven of the nine blocks mix signs across families (VOC/YOLO11n is uniformly positive, VOC/YOLO11m uniformly negative)—the heterogeneity quantified in the text.

same 128-image calibration budget and changes only the preprocessing—KITTI J95 AP moves from 6.72 to 27.91, recovering 21.2 of the 41.8-point gap to the diagnostic FP8 build. On the frozen protocol’s original-clean inputs the historical engine reads 7.21 against the matched arm’s 28.29 (Table 4), a 42.3-point total deficit to matched FP8; that pair additionally differs in builder policy (the historical engine predates the study and carries TensorRT’s default TF32 state), so the diagnostic comparison is the attributional one. The legacy recipe’s excess over the matched arm is therefore a mismatched-calibration artifact, not evidence about INT8 per se; what remains after the contract repair is the subject of the next observation.

Second, the residual INT8-matched collapse is severe but dataset-dependent: -21.2 AP versus matched FP8 on KITTI but only -2.9 AP on VOC (original clean images,

Table 4; -21.1 and -3.0 on the J95 basis). The paired interaction  $\Delta E$  (FP8-INT8-matched, corrupt mean) is -4.69 [-5.49, -3.86] on KITTI and -0.26 [-0.58, +0.06] on VOC; the KITTI narrowing is consistent with a scale artifact of the additive estimand: INT8-matched enters corruption 21 points below FP8, and the two arms’ retention rates are nearly equal (0.72 versus 0.75), so a similar relative decline narrows the absolute gap mechanically.

Third, the decomposition localizes the collapse to one model region (Table 5, diagnostic single-point arms on KITTI): quantizing only the classification head preserves accuracy ( $50.57 \approx \text{FP32}$ ), quantizing only the regression head reproduces the collapse (16.41), and INT8 everywhere *except* the regression head recovers to 49.50—within 0.94 of the diagnostic FP32 build on the same J95 basis (50.44; the paired-protocol FP32 reference in Table 4, evaluated on



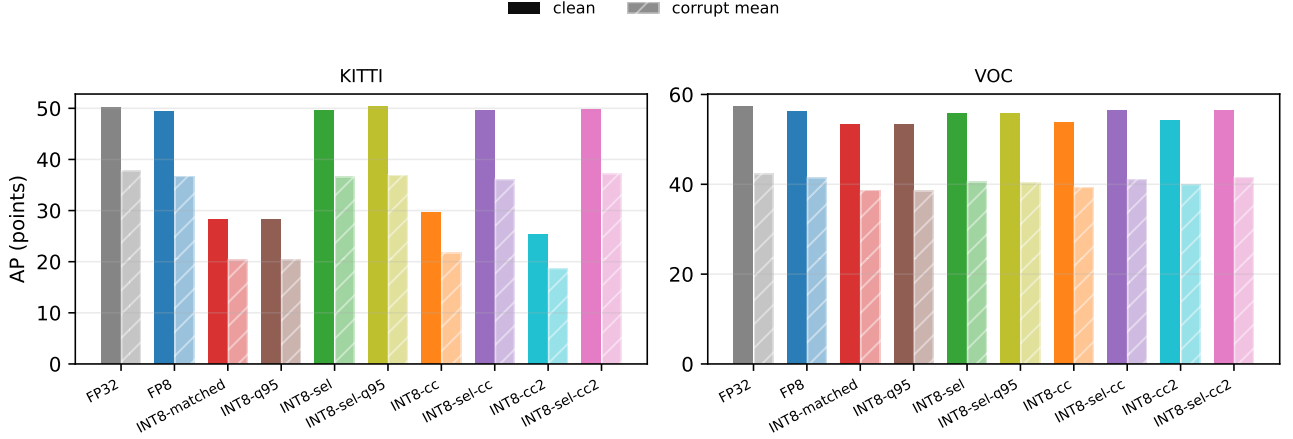
**Figure 5:** RetinaNet AP by corruption family and severity for the four matched-protocol arms on KITTI (top) and VOC (bottom). The “clean” point is the codec-control J95 cell for the matched arms; the FP32 engine predates codec control, so its baseline is the original-clean value (dashed). INT8-matched sits far below the other arms from the clean baseline onward on KITTI and visibly below them on most VOC cells (the VOC interaction itself is null; Section 4.2), while INT8-selective (regression head in floating point) tracks FP8-matched closely on every family; all four arms converge at the shared motion-blur-s5 failure wall.

original clean bytes rather than J95, is 50.15) and above the diagnostic full-graph FP8 build (48.50, at the 128-image diagnostic budget). Across all eleven INT8 diagnostic arms the outcome is bimodal in one variable: every arm that quantizes the regression head lands between 6.72 and 28.74 AP, whatever its calibration contract, estimator, budget, or site placement, and every arm that leaves it in floating point lands between 49.14 and 50.57 AP. Two alternatives a reader might raise are inconsistent with this split. Position in the graph is not the explanation, because the classification head is equally terminal yet harmless when quantized alone. Quantization coverage is not the explanation either: the restricted Conv+Add placement (15.75) quantizes fewer sites than full INT8 and does worse, while backbone+FPN quantization with both heads in floating point (49.14) quantizes most of the network and does well. Effects are non-additive (quantizing both heads, 17.09, is worse than quantizing the full graph), so the arms identify a dominant fragile component rather than additive contributions. These arms are single engine builds from one calibration draw on the development set; the frozen paired protocol validates the exclusion direction on both datasets, not the individual diagnostic values.

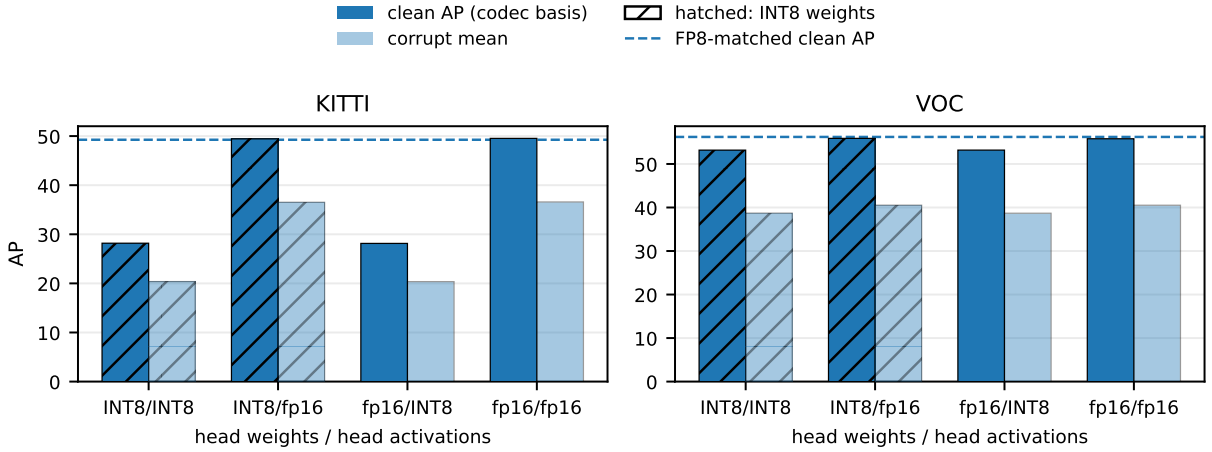
Fourth—the frozen-protocol check—on the paired protocol the selective arm validates the decomposition: INT8-selective matches FP8-matched within intervals that include zero on the clean control and on the 12-cell corrupt mean on KITTI (FP8 minus selective: J95 gap  $-0.28 [-1.11, +0.56]$ , corrupt-mean gap  $+0.17 [-0.07, +0.40]$ ; we note the J95 interval,  $-0.28 [-1.11, +0.56]$ , spans over an AP point around zero, so the match is a bound rather than tight equivalence). That match is a statement about the mean, not about every family: measured on the same  $\Delta E$  basis (family gap minus the J95 gap), the selective residual widens significantly under motion blur ( $+1.30 [+0.51, +2.04]$ ) and is null on JPEG ( $-0.11 [-0.81, +0.50]$ ), so family-level divergence remains. On VOC—the cleaner check—a small residual remains on both bases: FP8-matched minus selective is  $+0.42 [+0.13, +0.74]$  on the J95 clean control and  $+0.96 [+0.83, +1.09]$  on the corrupt mean, widening under corruption ( $\Delta E +0.54 [+0.23, +0.82]$ ). The regression head is therefore the dominant contributor, but not the entire mechanism on VOC.

Fifth, the operand factorial (Section 3.7; Fig. 7, Table 6) pins the deficit to a single operand class. Restoring only the head’s *activations* to fp16 while leaving head weights in INT8 recovers the collapse almost exactly: J95 rises to 49.44 AP on KITTI and 55.88 on VOC (100% and

## Recipe determinants of quantization fragility under corruption



**Figure 6:** RetinaNet recipe arms on KITTI and VOC: clean AP versus 12-cell corruption mean. INT8-matched collapses on KITTI (dataset-dependent); selective precision (INT8-sel = INT8-selective, reg-head in floating point) restores FP8-level behavior on KITTI and leaves a small residual on VOC; the calibration variants (INT8-q95 = codec-only calibration; INT8-cc = fold-1 corruption-aware calibration on noise and JPEG; INT8-cc2 = fold-2 calibration on fog and motion blur; a "sel-" prefix marks the selective recipe with the regression head in floating point) shift the matched arm little.



**Figure 7:** Operand-level factorial inside the RetinaNet regression head (post-hoc arms on the frozen paired conditions). Each group holds head weights and head activations at INT8 or fp16 independently; solid bars are codec-control clean AP and translucent bars the 12-cell corrupt mean. On both datasets the INT8-weights/fp16-activations arm matches the selective arm within 0.1 AP; on KITTI it also reaches the FP8-matched clean level, while on VOC both trail FP8-matched as the selective arm does. The fp16-weights/INT8-activations arm tracks the fully quantized matched arm almost exactly: the deficit follows head *activation* quantization, with a residual weight effect bounded near 0.2 AP on KITTI.

103% of the matched-to-selective gap respectively), and the residual to full head exclusion is sub-0.2-AP in magnitude (selective minus W-only +0.09 [+0.01, +0.17] on KITTI, -0.09 [-0.20, +0.02] on VOC). The complementary arm—fp16 head weights over INT8 head activations—changes nothing: a clean-level contrast to the matched arm of -0.04 [-0.20, +0.11] on KITTI and +0.01 [-0.08, +0.10] on VOC, both intervals enclosing zero, and the same null holds on the interaction basis ( $\Delta E$  +0.03 [-0.14, +0.20] KITTI, -0.03 [-0.13, +0.06] VOC). The head localization is therefore sharper than a region statement: within

this recipe the failure is overwhelmingly an *activation*-quantization phenomenon of the regression head—head weights in INT8 contribute at most a residual penalty bounded at roughly 0.2 AP (the W-only contrast above is the weight effect when activations are already fp16, and the A-only contrast is the weight effect when they are not; both are within 0.1 AP of zero in point estimate, though the KITTI selective-minus-W-only interval just excludes it). The factorial interaction is correspondingly compatible with zero on the clean basis (+0.14 [-0.04, +0.30] KITTI, -0.09 [-0.23, +0.05] VOC), while the corrupt-mean interaction is a small positive point on KITTI

**Table 4**

RetinaNet arms under the frozen protocol (AP points, COCO-style AP on converted annotations). Corrupt mean = equal weight over the 12 corruption-severity cells. J95: codec-control clean; FP32 and legacy arms predate codec control (dashes). The INT8-legacy arm is the pre-study historical engine and additionally differs from the matched arms in builder policy (TensorRT-default TF32 state versus TF32 disabled in the TensorRT Python builder); the controlled preprocessing-only contrast is the diagnostic rebuild in Table 5.

| Dataset | Arm                             | Clean AP | J95 AP | Corrupt mean AP |
|---------|---------------------------------|----------|--------|-----------------|
| KITTI   | FP32 reference                  | 50.15    | —      | 37.80           |
|         | FP8-matched                     | 49.48    | 49.25  | 36.75           |
|         | INT8-legacy (contract mismatch) | 7.21     | —      | 7.43            |
|         | INT8-matched                    | 28.29    | 28.18  | 20.37           |
|         | INT8-q95calib (codec control)   | 28.26    | 27.85  | 20.42           |
|         | INT8-selective (reg-head FP32)  | 49.60    | 49.53  | 36.58           |
|         | INT8-sel-q95calib               | 50.28    | 49.79  | 36.94           |
|         | INT8-corruptcalib               | 29.69    | 29.74  | 21.71           |
|         | INT8-sel-corruptcalib           | 49.55    | 49.03  | 36.08           |
|         | INT8-cc2calib (fold 2)          | 25.36    | 25.47  | 18.68           |
|         | INT8-sel-cc2calib (fold 2)      | 49.72    | 50.54  | 37.20           |
|         |                                 |          |        |                 |
| VOC     | FP32 reference                  | 57.23    | —      | 42.32           |
|         | FP8-matched                     | 56.17    | 56.21  | 41.48           |
|         | INT8-legacy (contract mismatch) | 33.61    | —      | 24.32           |
|         | INT8-matched                    | 53.30    | 53.18  | 38.71           |
|         | INT8-q95calib (codec control)   | 53.33    | 53.39  | 38.54           |
|         | INT8-selective (reg-head FP32)  | 55.87    | 55.79  | 40.52           |
|         | INT8-sel-q95calib               | 55.81    | 55.95  | 40.40           |
|         | INT8-corruptcalib               | 53.78    | 53.91  | 39.35           |
|         | INT8-sel-corruptcalib           | 56.36    | 56.43  | 41.13           |
|         | INT8-cc2calib (fold 2)          | 54.15    | 54.22  | 39.97           |
|         | INT8-sel-cc2calib (fold 2)      | 56.40    | 56.43  | 41.58           |
|         |                                 |          |        |                 |

**Table 5**

Diagnostic recipe decomposition on KITTI (single-point AP on the J95 codec-control inputs, 512-image calibration unless stated). These arms guided the paired-protocol design and carry no intervals; the “INT8 matched preprocessing, 512-img calibration” row reproduces the paired INT8-matched cell’s numbers from the same quantized ONNX graph (identical predictions; the serialized engine is a separate build, so the engine hash differs) rather than duplicating it.

| Recipe arm (KITTI diagnostic, single-point)            | J95 AP |
|--------------------------------------------------------|--------|
| INT8 legacy (mismatched calibration contract, 128-img) | 6.72   |
| INT8 matched preprocessing, 128-img calibration        | 27.91  |
| INT8 matched preprocessing, 512-img calibration        | 28.18  |
| INT8 max-estimator, 128-img calibration                | 23.94  |
| INT8 restricted to Conv+Add sites, 128-img calibration | 15.75  |
| INT8 at exactly the FP8 compute sites (shared mask)    | 28.74  |
| INT8 backbone+FPN only (heads FP32)                    | 49.14  |
| INT8 classification head only                          | 50.57  |
| INT8 regression head only                              | 16.41  |
| INT8 heads only (cls+reg)                              | 17.09  |
| INT8 all-except regression head                        | 49.50  |
| FP8 matched, 128-img calibration                       | 48.50  |
| FP32 reference (diagnostic build)                      | 50.44  |

(+0.07 [+0.01, +0.14], unadjusted) and compatible with zero on VOC (+0.01 [−0.03, +0.05])—so the two operands act nearly additively, with at most a 0.1-AP-scale interaction on the corrupted inputs, and the weight contribution stays within the bounds already given. This operand attribution is also consistent with why matched FP8 does not fail—its E4M3 exponent range tolerates the head’s activation tails—though whether the operative INT8 cost is range clipping or bin resolution is not separated by this design,

**Table 6**

The 2x2 factorial of Table 4’s regression-head exclusion, decomposed by operand (post-hoc arms; J95 intervals vs the INT8/INT8 matched arm at 95%, paired bootstrap). “J95 vs W8A8” is the clean-level contrast to the matched arm. Non-quantized operands compute in fp16 under the INT8 engine’s fp16 fallback.

| Dataset | Head weights | Head acts. | J95   | Corrupt mean | Retention | J95 vs W8A8              |
|---------|--------------|------------|-------|--------------|-----------|--------------------------|
| KITTI   | INT8         | INT8       | 28.18 | 20.37        | 0.72      | —                        |
|         | INT8         | fp16       | 49.44 | 36.52        | 0.74      | +21.25* [+19.88, +22.69] |
|         | fp16         | INT8       | 28.14 | 20.36        | 0.72      | −0.04 [−0.20, +0.11]     |
|         | fp16         | fp16       | 49.53 | 36.58        | 0.74      | +21.35* [+19.95, +22.81] |
| VOC     | INT8         | INT8       | 53.18 | 38.71        | 0.73      | —                        |
|         | INT8         | fp16       | 55.88 | 40.53        | 0.73      | +2.70* [+2.46, +2.96]    |
|         | fp16         | INT8       | 53.19 | 38.68        | 0.73      | +0.01 [−0.08, +0.10]     |
|         | fp16         | fp16       | 55.79 | 40.52        | 0.73      | +2.61* [+2.37, +2.88]    |

and the FP8 comparison quantizes both operands so it does not isolate the activation pathway on its own.

A corruption-severity boundary is also shared across precisions (Fig. 5): at motion-blur severity 5 all matched-protocol arms drop to below 6 AP on both datasets—KITTI: FP32 5.45, FP8-matched 5.59, INT8-matched 3.75, selective 4.52; VOC: 5.71/5.54/4.89/4.88 in the same order—FP32 included. That cell measures a common failure wall, not an INT8-specific deficit, and its equal weight in the corrupt mean compresses all contrasts built on it; the leave-one-out sensitivity removes it without changing any frozen-criterion verdict (two secondary codec-control contrasts do move across the zero boundary on VOC; Supplement S8).

### 4.3. An anchor-free detector bounds the mechanism’s generality

The second-architecture check (Section 3.7; Table 7)—a third detector family in the study, all intervals unadjusted (Section 5.4)—asks whether the activation-fragile regression head is a property of INT8 detectors generally. It is not: on anchor-free FCOS the INT8-matched arm develops no deficit at the RetinaNet scale—J95 clean-control AP is 47.53 on KITTI and 49.02 on VOC, against 47.60 and 49.22 for the FP32 reference (12-cell corrupt means 33.82/33.95 KITTI, 34.59/34.88 VOC). Small INT8-to-FP32 gaps do remain: the FP32-minus-selective contrast is compatible with zero on the KITTI clean basis (+0.07 [−0.25, +0.42]) but excludes zero on VOC clean (+0.19 [+0.04, +0.34]) and on both corrupt means (+0.12 [+0.01, +0.22]/+0.30 [+0.26, +0.36]). Those gaps are smaller than the one FCOS’s own FP8-matched arm leaves to FP32 in point estimate (clean 47.60 vs. 47.06 KITTI, 49.22 vs. 48.88 VOC)—and they are not head-specific: head exclusion changes nothing (selective minus matched J95 0.00 [−0.15, +0.12] KITTI, +0.01 [−0.04, +0.08] VOC; corrupt mean 0.00 [−0.06, +0.04]/−0.01 [−0.030, +0.005]), so there is no regression-head INT8 cost for the exclusion to repair. If anything the direction tilts the other way: the matched FP8–INT8 clean contrast is a small negative point estimate on both datasets—compatible with zero on VOC (−0.14 [−0.33, +0.07]) and marginally negative on KITTI (−0.47 [−0.990, −0.001], a borderline unadjusted

**Table 7**

Cross-architecture check (post-hoc arm set): the RetinaNet regression-head deficit does not replicate on FCOS. Paired-bootstrap contrasts at 95% under the frozen conditions; on FCOS the selective-minus-matched J95 contrast includes zero on both datasets—there is no head-specific INT8 cost for the exclusion to repair. Detector rows are separate bootstrap blocks; no cross-detector contrast is estimated.

| Detector  | Dataset | FP8-INT8 J95                         | FP8-INT8 $\Delta E$               | Sel-INT8 J95                         | Sel-INT8 $\Delta E$               | FP8-Sel J95                       |
|-----------|---------|--------------------------------------|-----------------------------------|--------------------------------------|-----------------------------------|-----------------------------------|
| RetinaNet | KITTI   | +21.07 <sup>†</sup> [+19.57, +22.61] | -4.69 <sup>†</sup> [-5.49, -3.86] | +21.35 <sup>†</sup> [+19.95, +22.81] | -5.14 <sup>†</sup> [-5.94, -4.34] | -0.28 [-1.11, +0.56]              |
| FCOS      | KITTI   | -0.47 <sup>†</sup> [-0.990, -0.001]  | -0.04 [-0.44, +0.40]              | 0.00 [-0.15, +0.12]                  | 0.00 [-0.13, +0.15]               | -0.47 [-0.99, +0.05]              |
| RetinaNet | VOC     | +3.03 <sup>†</sup> [+2.66, +3.41]    | -0.26 [-0.58, +0.06]              | +2.61 <sup>†</sup> [+2.37, +2.88]    | -0.80 <sup>†</sup> [-0.98, -0.61] | +0.42 <sup>†</sup> [+0.13, +0.74] |
| FCOS      | VOC     | -0.14 [-0.33, +0.07]                 | +0.07 [-0.13, +0.25]              | +0.01 [-0.04, +0.08]                 | -0.03 [-0.09, +0.03]              | -0.16 [-0.36, +0.05]              |

interval)—and the corrupt-mean contrast is negative on both ( $-0.50$  [ $-0.66, -0.36$ ] KITTI,  $-0.08$  [ $-0.15, -0.01$ ] VOC), opposite in sign to RetinaNet’s KITTI matched deficit of  $>21$  AP. The corruption-mean interaction is compatible with zero (FP8-matched minus INT8-matched  $\Delta E$   $-0.04$  [ $-0.44, +0.40$ ] KITTI,  $+0.07$  [ $-0.13, +0.25$ ] VOC), though—as on RetinaNet—the mean statement does not extend to every family: on VOC the JPEG and motion-blur family  $\Delta E$  intervals exclude zero. The RetinaNet fragility therefore does not transfer to a differently parametrized head—FCOS regresses per-pixel distances plus a centerness term rather than anchor deltas—which is consistent with recipe-dependence: the collapse tracks a particular head under a particular recipe, not the INT8 format. Two caveats bound the claim: FCOS differs from RetinaNet in decode parametrization, anchor density, and centerness scoring (and its selective arm additionally dequantizes centerness, so the treatments are not identical), so the non-replication speaks to recipe-dependence, not to anchor-based heads as a class; and all four RetinaNet/FCOS checkpoints (one per detector×dataset) are early minimum-validation-loss selections (run-log epochs 21/5 RetinaNet KITTI/VOC, 14/10 FCOS KITTI/VOC of a 100-epoch schedule; the VOC FCOS training-loss divergence at epoch 70 postdates its epoch-10 selection point and did not affect it), so differing (uncontrolled) training state—not precision—is a live alternative explanation for the absent deficit.

#### 4.4. Corruption-aware calibration is itself context-dependent

If the INT8-matched arm’s corrupted-input accuracy were limited by clean-only calibration ranges, the corrupt-calibration intervention should recover held-out corruption AP. Against the frozen criterion, the result splits by dataset (Table 8, Fig. 8). On VOC the held-out difference is  $+0.02$  [ $-0.13, +0.16$ ] AP, null, while the in-family cells seen in calibration improve by  $+1.27$  [ $+1.12, +1.42$ ]. On KITTI the primary criterion is met—held-out  $+1.07$  [ $+0.87, +1.26$ —but the gain closes less than a tenth of the held-out residual to FP8-matched on the same cells (INT8-corruptcalib minus FP8-matched:  $-13.34$  [ $-14.13, -12.57$ ]). Excluding the saturated motion-blur-s5 cell leaves both verdicts unchanged (held-out  $+1.27$  [ $+1.06, +1.50$ ] on KITTI;  $+0.07$  [ $-0.10, +0.23$ ] on VOC).

The paper’s own estimand explains why the KITTI gain is not demonstrably corruption-specific. Corrupt calibration also raises *clean-control* AP—by  $+1.56$  [ $+1.02, +2.10$ ] on KITTI and  $+0.73$  [ $+0.40, +1.01$ ] on VOC—so under the additive parallel-shift assumption of Section 3.6, part of the corrupted-cell gains would appear as a uniform accuracy shift unrelated to corruption. Subtracting that component, the corruption-specific held-out effect  $\Delta E_{\text{held-out}}$  is  $-0.50$  [ $-1.01, +0.03$ ] on KITTI and  $-0.71$  [ $-1.02, -0.39$ ] on VOC: null-to-negative on both datasets. The KITTI held-out gain is therefore not demonstrably larger than the gain on uncorrupted images. Only the calibrated families show a corruption-specific benefit, and only on VOC ( $\Delta E_{\text{in-family}}$   $+0.54$  [ $+0.24, +0.88$ ]; KITTI  $+0.05$  [ $-0.54, +0.59$ ]). Calibration on corruptions fits the calibrated distribution and shifts overall accuracy slightly; in this fold it does not transfer robustness to unseen corruption families—whether that verdict generalizes is what the complementary fold tests below.

On the selective arm the pattern repeats at smaller magnitude. On VOC, corrupt calibration adds  $+0.61$  [ $+0.49, +0.72$ ] AP on the corrupt mean, driven by the in-family cells ( $+1.23$  [ $+1.08, +1.39$ ]) with a null held-out difference ( $-0.01$  [ $-0.17, +0.13$ ]), and improves the signed corrupt-mean residual (selective minus FP8-matched) from  $-0.96$  to  $-0.35$  [ $-0.46, -0.24$ ] without closing it; the held-out residual is  $-0.51$  [ $-0.66, -0.38$ ]. The second frozen criterion—held-out residual to FP8-matched including zero—is therefore not met on either dataset (VOC  $-0.51$  [ $-0.66, -0.38$ ]; KITTI  $-1.30$  [ $-1.61, -1.00$ ]): calibration does not explain the residual. The fold-1 primary estimand also covers the selective-tier held-out contrast itself (sel-corruptcalib minus sel), which is  $-0.58$  [ $-0.80, -0.36$ ] on KITTI and  $-0.01$  [ $-0.17, +0.13$ ] on VOC—significantly negative on KITTI, under the contamination caveat of Section 3.4.

The complementary fold reverses the family roles (Fig. 2), and its result is not a second null: the intervention’s own effect is itself sign-unstable across the four fold×dataset cells (Table 8, Fig. 8). On KITTI the fold-2 arm is significantly *worse* on every level basis (J95 clean  $-2.71$  [ $-3.38, -1.93$ ]; calibrated fog+motion-blur cells  $-1.35$  [ $-1.56, -1.04$ ]; held-out  $-2.02$  [ $-2.31, -1.73$ ]): the held-out change meets the literal nonzero-CI wording of the fold-2 rule but is a harm, so the directional reading—and the replication it was meant to show—fails, and the clean-level shift flips sign relative to fold 1. The degradation is largest on the clean control, so the corruption-specific components are positive on the calibrated families ( $\Delta E_{\text{in-family}}$   $+1.36$  [ $+0.61, +2.16$ ]) and null held-out ( $+0.69$  [ $-0.10, +1.37$ ]; codec-anchored  $+0.37$  [ $-0.28, +0.90$ ]): fold 2 damages this engine broadly rather than improving any level basis.

On VOC the same fold-2 recipe meets the primary rule: held-out  $+1.66$  [ $+1.52, +1.83$ —recovering roughly half of the held-out residual to FP8-matched on those families ( $-1.74$  [ $-1.94, -1.53$ ] remaining)—and, uniquely among

the four cells, its corruption-specific component is positive and significant ( $\Delta E_{\text{held-out}} +0.62$  [ $+0.30, +0.96$ ], confirmed against the codec anchor at  $+1.06$  [ $+0.76, +1.38$ ]).

The selective tier does not repeat the split but diverges from it (all selective fold-2 contrasts are post-hoc; the frozen fold-2 rules cover only the full tier): on KITTI the fold-2 selective arm is significantly *lifted* where the full-INT8 arm is harmed (J95  $+1.01$  [ $+0.37, +1.72$ ]; held-out  $+0.66$  [ $+0.33, +0.92$ ];  $\Delta E$  null; same development-contamination caveat as the fold-1 selective arm)—although neither lift is significant against its codec-matched baseline (sel-cc2 minus sel-q95, J95  $+0.75$  [ $-0.04, +1.56$ ]; held-out  $+0.25$  [ $-0.09, +0.52$ ]), so codec adaptation can account for this particular lift. On VOC the selective held-out gain is  $+1.50$  [ $+1.35, +1.66$ ] and does survive its codec anchor ( $+1.64$  [ $+1.49, +1.80$ ]), with a corruption-specific component whose point estimate exceeds the full tier’s ( $\Delta E_{\text{held-out}} +0.86$  [ $+0.55, +1.18$ ], codec-anchored  $+1.16$  [ $+0.87, +1.49$ ]; no between-tier interval is computed)—against FP8-matched its held-out residual (INT8-sel-cc2calib minus FP8-matched) is  $+0.08$  [ $-0.06, +0.23$ ], the only cell in which the fold-1 analog of the second criterion (an interval including zero) is met; on KITTI fold 2 the same contrast excludes zero in the favorable direction ( $+1.04$  [ $+0.64, +1.40$ ], the selective fold-2 arm significantly exceeding FP8-matched on held-out, under the contamination caveat).

Corruption-aware calibration therefore *can* transfer, but whether it does depends on the dataset $\times$ family partition: across the two folds its primary criterion is met on KITTI fold 1 and VOC fold 2 and fails on the other two cells, including a significant harm on KITTI fold 2.

Two codec-matched control arms separate codec adaptation from corruption coverage across both folds. Corrupt-calibration images carry a JPEG quality-95 terminal encoding that the clean-calibration images lack, so part of any gain could reflect codec adaptation rather than corruption coverage. Calibrating the same 512 images at quality 95 without corruption (INT8-q95calib) produces no significant clean-AP change on either dataset (J95 clean  $-0.34$  [ $-0.93, +0.35$ ] on KITTI,  $+0.20$  [ $-0.07, +0.48$ ] on VOC) and does not improve the corrupt mean ( $+0.04$  [ $-0.10, +0.22$ ] on KITTI;  $-0.17$  [ $-0.26, -0.09$ ] on VOC, a small significant reduction). Against this anchor, fold 1’s corrupt-calibration contrast is essentially unchanged in level terms (held-out  $+0.95$  [ $+0.73, +1.15$ ] on KITTI and  $+0.13$  [ $-0.02, +0.28$ ] on VOC), and its corruption-specific held-out component is significantly negative on both datasets ( $\Delta E_{\text{held-out}} -0.94$  [ $-1.60, -0.33$ ] on KITTI;  $-0.39$  [ $-0.69, -0.07$ ] on VOC); on the selective tier, codec matching alone produces a small shift on corrupted cells that flips sign across datasets (corrupt mean  $+0.36$  [ $+0.23, +0.49$ ] on KITTI;  $-0.12$  [ $-0.21, -0.04$ ] on VOC) with no significant held-out corruption-specific component ( $+0.05$  [ $-0.48, +0.57$ ] and  $-0.27$  [ $-0.56, +0.02$ ]), though the codec arm’s own in-family interaction is itself significantly negative on VOC

( $-0.30$  [ $-0.60, -0.02$ ]). Codec effects are therefore real but small in *level* terms—aggregate point estimates within half an AP point, far below the 21.2-AP KITTI level deficit and the intervention level shifts (at most 2.7 AP)—and explain neither the collapse nor the level gains. On the interaction basis, however, codec effects are the same order as the corruption-specific components themselves (VOC q95-calib  $\Delta E_{\text{held-out}} -0.31$  [ $-0.62, -0.02$ ]), and switching the anchor moves VOC fold-2’s  $\Delta E_{\text{held-out}}$  from  $+0.62$  [ $+0.30, +0.96$ ] to  $+1.06$  [ $+0.76, +1.38$ ]; the codec-anchored  $\Delta E$  rows in Table 8 are therefore the relevant basis for the corruption-specific verdict, not an optional robustness check.

The additive reading is itself scale-sensitive (Section 4.2), so a proportional-shift check disciplines it: scaling each fold’s clean-control change by the matched arm’s retention on that fold’s held-out families—0.63 for fold 1 and 0.82 for fold 2 on KITTI (0.67 and 0.78 on VOC)—predicts held-out changes of  $+0.98$  and  $-2.22$  on KITTI and  $+0.49$  and  $+0.82$  on VOC, against the observed  $+1.07, -2.02, +0.02$ , and  $+1.66$ . Both KITTI cells are therefore largely scale effects; VOC fold 1 shows no corruption-specific benefit under either reading (observed  $+0.02$  against a retention-scaled prediction of  $+0.49$ ); only VOC fold 2 exceeds its prediction materially ( $+1.66$  vs.  $+0.82$ ), so that cell’s positive component is not an artifact of the additive basis. The check anchors on INT8-matched; the codec anchor gives the same qualitative answer (KITTI fold-1 residual  $-0.23$  AP).

The intervention verdict must be stated in two parts, separating the frozen rules from the interpretive analyses (Table 1). Under the frozen rules alone, the primary held-out test succeeds on KITTI fold 1 and VOC fold 2; on VOC fold 1 it is null, and on KITTI fold 2 the change meets the literal nonzero-CI wording but is a significant *harm*, so the directional reading fails; fold 1’s null-seeking selective test fails on both datasets; and the fold-2 frozen corruption-specific rule is met on VOC and not met on KITTI (the fold-2 selective-versus-FP8 contrast is a post-hoc analog, met on VOC and exceeded favorably on KITTI under the contamination caveat; the frozen selective-tier held-out estimand, which carries no decision rule, is significantly negative on KITTI ( $-0.58$  [ $-0.80, -0.36$ ]) and null on VOC ( $-0.01$  [ $-0.17, +0.13$ ])). Under the  $\Delta E_{\text{held-out}}$  decomposition—post hoc for fold 1, a frozen rule for fold 2—fold 1 shows no corruption-specific benefit on either dataset, while fold 2 produces a genuine positive component on VOC—and codec adaptation cannot account for either direction of that instability. Two conclusions follow, and they should be kept apart. First, corruption-aware entropy calibration is itself context-dependent: it transfers a real corruption-specific benefit in one of four fold $\times$ dataset cells, leaves the others null or harmful, and never approaches the scale of the KITTI level deficit—a second instance of the recipe $\times$ context dependence the paired layer showed. Second, on KITTI, where the level deficit lives, the tested interventions do not repair it in level

**Table 8**

RetinaNet intervention and codec-control contrasts (A minus B, AP points; plug-in point with 95% paired-bootstrap interval, 2,000 common-image resamples). J95 clean: codec-control clean cell; corrupt mean: 12 cells; in-family: Gaussian noise and JPEG (the fold-1 calibrated families); held-out: fog and motion blur (never calibrated in fold 1); for the q95-calibration rows, which see clean images only, these labels mark the evaluation partition rather than exposure;  $\Delta E_{\text{held-out}}$  = held-out difference minus J95 clean difference, and  $\Delta E_{\text{in-family}}$  is defined analogously; because every row here is an intervention-minus-baseline contrast (Eq. (3)), positive  $\Delta E$  shifts the difference toward the first-named arm. For rows anchored to a clean-calibrated INT8 arm, a positive  $\Delta E$  is a corruption-specific intervention benefit; for rows anchored to FP8-matched—which remain deep level deficits—positive values describe gap narrowing under corruption, not a benefit. This ordering is opposite to the FP8-INT8  $\Delta E$  of Table 3. Fold-2 rows swap the family roles: their in-family is fog and motion blur and their held-out is Gaussian noise and JPEG. \* interval excludes zero.

| Contrast (A – B)                               | Dataset | J95 clean                | Corrupt mean             | In-family                | Held-out                 | $\Delta E_{\text{in-family}}$ | $\Delta E_{\text{held-out}}$ |
|------------------------------------------------|---------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------------|------------------------------|
| INT8-corruptcalib – INT8-matched               | KITTI   | +1.56* [+1.02, +2.10]    | +1.34* [+1.17, +1.48]    | +1.61* [+1.37, +1.79]    | +1.07* [+0.87, +1.26]    | +0.05 [–0.54, +0.59]          | –0.50 [–1.01, +0.03]         |
| INT8-corruptcalib – INT8-matched               | VOC     | +0.73* [+0.40, +1.01]    | +0.64* [+0.52, +0.76]    | +1.27* [+1.12, +1.42]    | +0.54* [+0.24, +0.88]    | +0.34* [+0.24, +0.44]         | –0.71* [–1.02, –0.39]        |
| INT8-q95calib – INT8-matched                   | KITTI   | –0.34 [–0.93, +0.35]     | +0.04 [–0.10, +0.22]     | –0.02 [–0.23, +0.21]     | +0.11 [–0.05, +0.33]     | +0.31 [–0.35, +0.90]          | +0.45 [–0.24, +1.04]         |
| INT8-q95calib – INT8-matched                   | VOC     | +0.20 [–0.07, +0.48]     | –0.17* [–0.26, –0.09]    | –0.23* [–0.37, –0.11]    | –0.11* [–0.228, –0.002]  | –0.44* [–0.73, –0.14]         | –0.31* [–0.62, –0.02]        |
| INT8-corruptcalib – INT8-q95calib              | KITTI   | +1.90* [+1.28, +2.54]    | +1.29* [+1.08, +1.45]    | +1.64* [+1.33, +1.86]    | +0.95* [+0.73, +1.15]    | –0.26 [–0.89, +0.30]          | –0.94* [–1.60, –0.33]        |
| INT8-corruptcalib – INT8-q95calib              | VOC     | +0.52* [+0.22, +0.80]    | +0.81* [+0.70, +0.93]    | +1.50* [+1.35, +1.66]    | +0.13 [–0.02, +0.28]     | +0.98* [+0.70, +1.31]         | –0.39* [–0.69, –0.07]        |
| INT8-sel-corruptcalib – INT8-selective         | KITTI   | –0.50 [–0.96, +0.03]     | –0.50* [–0.65, –0.36]    | –0.41* [–0.63, –0.24]    | –0.58* [–0.80, –0.36]    | +0.09 [–0.43, +0.52]          | –0.08 [–0.68, +0.41]         |
| INT8-sel-corruptcalib – INT8-selective         | VOC     | +0.63* [+0.33, +0.94]    | +0.61* [+0.49, +0.72]    | +1.23* [+1.08, +1.39]    | +0.60* [+0.29, +0.92]    | +0.60* [+0.29, +0.92]         | –0.65* [–0.97, –0.35]        |
| INT8-sel-q95calib – INT8-selective             | KITTI   | +0.26 [–0.15, +0.74]     | +0.36* [+0.23, +0.49]    | +0.41* [+0.21, +0.61]    | +0.31* [+0.13, +0.54]    | +0.15 [–0.39, +0.58]          | +0.05 [–0.48, +0.57]         |
| INT8-sel-q95calib – INT8-selective             | VOC     | +0.16 [–0.10, +0.43]     | –0.12* [–0.21, –0.04]    | –0.14* [–0.27, –0.01]    | –0.11 [–0.22, +0.01]     | –0.30* [–0.60, –0.02]         | –0.27 [–0.56, +0.02]         |
| INT8-sel-corruptcalib – INT8-sel-q95calib      | KITTI   | –0.76* [–1.35, –0.17]    | –0.85* [–1.05, –0.69]    | –0.82* [–1.08, –0.59]    | –0.89* [–1.16, –0.65]    | –0.06 [–0.65, +0.54]          | –0.13 [–0.83, +0.51]         |
| INT8-sel-corruptcalib – INT8-sel-q95calib      | VOC     | +0.47* [+0.16, +0.76]    | +0.73* [+0.61, +0.85]    | +1.37* [+1.22, +1.54]    | +0.09 [–0.07, +0.25]     | +0.90* [+0.61, +1.22]         | –0.38* [–0.68, –0.08]        |
| INT8-sel-corruptcalib – FP8-matched            | KITTI   | –0.23 [–1.12, +0.86]     | –0.67* [–0.91, –0.44]    | –0.03 [–0.37, +0.31]     | –1.30* [–1.61, –1.00]    | +0.20 [–0.74, +0.92]          | –1.08* [–2.18, –0.16]        |
| INT8-sel-corruptcalib – FP8-matched            | VOC     | +0.21 [–0.06, +0.49]     | –0.35* [–0.46, –0.24]    | –0.19* [–0.32, –0.05]    | –0.51* [–0.66, –0.38]    | –0.40* [–0.67, –0.12]         | –0.72* [–1.00, –0.44]        |
| INT8-corruptcalib – FP8-matched                | KITTI   | –19.51* [–20.90, –18.04] | –15.04* [–15.96, –14.17] | –16.74* [–17.85, –15.66] | –13.34* [–14.13, –12.57] | +2.77* [+1.98, +3.48]         | +6.17* [+5.12, +7.10]        |
| INT8-corruptcalib – FP8-matched                | VOC     | –2.31* [–2.68, –1.98]    | –2.13* [–2.30, –1.97]    | –2.13* [–2.33, –1.93]    | –2.12* [–2.32, –1.96]    | +0.17 [–0.10, +0.49]          | +0.18 [–0.12, +0.50]         |
| INT8-cc2calib – INT8-matched (fold 2)          | KITTI   | –2.71* [–3.38, –1.93]    | –1.69* [–1.88, –1.44]    | –1.35* [–1.56, –1.04]    | –2.02* [–2.31, –1.73]    | +1.36* [+0.61, +2.16]         | +0.69 [–0.10, +1.37]         |
| INT8-cc2calib – INT8-matched (fold 2)          | VOC     | +1.04* [+0.72, +1.34]    | +1.26* [+1.15, +1.37]    | +0.86* [+0.72, +0.99]    | +1.66* [+1.52, +1.83]    | –0.19 [–0.49, +0.13]          | +0.62* [+0.30, +0.96]        |
| INT8-cc2calib – INT8-q95calib (fold 2)         | KITTI   | –2.37* [–2.90, –1.74]    | –1.73* [–1.96, –1.47]    | –1.47* [–1.68, –1.18]    | –2.00* [–2.32, –1.69]    | +0.91* [+0.28, +1.47]         | +0.37 [–0.28, +0.90]         |
| INT8-cc2calib – INT8-q95calib (fold 2)         | VOC     | +0.84* [+0.54, +1.12]    | +1.43* [+1.32, +1.55]    | +0.97* [+0.82, +1.12]    | +1.89* [+1.75, +2.06]    | +0.13 [–0.15, +0.43]          | +1.06* [+0.76, +1.38]        |
| INT8-sel-cc2calib – INT8-selective (fold 2)    | KITTI   | +1.01* [+0.37, +1.72]    | +0.62* [+0.42, +0.83]    | +0.59* [+0.39, +0.86]    | +0.66* [+0.33, +0.92]    | –0.42 [–1.08, +0.24]          | –0.35 [–1.09, +0.28]         |
| INT8-sel-cc2calib – INT8-selective (fold 2)    | VOC     | +0.63* [+0.34, +0.94]    | +1.06* [+0.96, +1.18]    | +0.63* [+0.48, +0.76]    | +1.50* [+1.35, +1.66]    | –0.01 [–0.33, +0.29]          | +0.86* [+0.55, +1.18]        |
| INT8-sel-cc2calib – INT8-sel-q95calib (fold 2) | KITTI   | +0.75 [–0.04, +1.56]     | +0.27* [+0.08, +0.43]    | +0.28* [+0.03, +0.52]    | +0.25 [–0.09, +0.52]     | –0.47 [–1.25, +0.33]          | –0.50 [–1.27, +0.33]         |
| INT8-sel-cc2calib – INT8-sel-q95calib (fold 2) | VOC     | +0.48* [+0.18, +0.75]    | +1.19* [+1.07, +1.30]    | +0.74* [+0.59, +0.87]    | +1.64* [+1.49, +1.80]    | +0.26 [–0.03, +0.56]          | +1.16* [+0.87, +1.49]        |
| INT8-sel-cc2calib – FP8-matched (fold 2)       | KITTI   | +1.29* [+0.64, +2.03]    | +0.45* [+0.20, +0.71]    | –0.13 [–0.44, +0.20]     | +1.04* [+0.64, +1.40]    | –1.42* [–2.14, –0.76]         | –0.25 [–0.93, +0.36]         |
| INT8-sel-cc2calib – FP8-matched (fold 2)       | VOC     | +0.22 [–0.06, +0.48]     | +0.11 [–0.01, +0.22]     | +0.13 [–0.02, +0.26]     | +0.08 [–0.06, +0.23]     | –0.08 [–0.36, +0.20]          | –0.14 [–0.40, +0.15]         |
| INT8-cc2calib – FP8-matched (fold 2)           | KITTI   | –23.78* [–25.30, –22.13] | –18.07* [–19.07, –17.00] | –15.76* [–16.63, –14.86] | –20.37* [–21.58, –19.14] | +8.02* [+7.05, +8.94]         | +3.41* [+2.51, +4.23]        |
| INT8-cc2calib – FP8-matched (fold 2)           | VOC     | –1.99* [–2.34, –1.66]    | –1.51* [–1.68, –1.35]    | –1.28* [–1.47, –1.12]    | –1.74* [–1.94, –1.53]    | +0.70* [+0.41, +1.01]         | +0.25 [–0.04, +0.57]         |

terms: fold 1’s raw held-out gain closes under a tenth of the residual, and fold 2 degrades every level basis including the families it calibrated on. The null-to-negative held-out interactions are a separate statement—they show that the gains which do occur are not corruption-specific, so the frozen hypothesis (that corrupted-input accuracy is limited by clean-only calibration ranges) is not supported—while the mechanism of the level deficit itself is not identified by either result. The contrast that removes the level deficit is instead the regression-head exclusion: it restores FP8-level behavior on KITTI and leaves only a small residual on VOC, without retraining. The level-deficit repair is also not a corruption-specific repair: the plain selective arm’s residual to FP8 still widens significantly on VOC and under KITTI motion blur (Section 4.2); only the post-hoc sel-cc2calib arm on VOC fold 2 attains a residual whose interval includes zero on every basis.

#### 4.5. Historical exploratory evidence

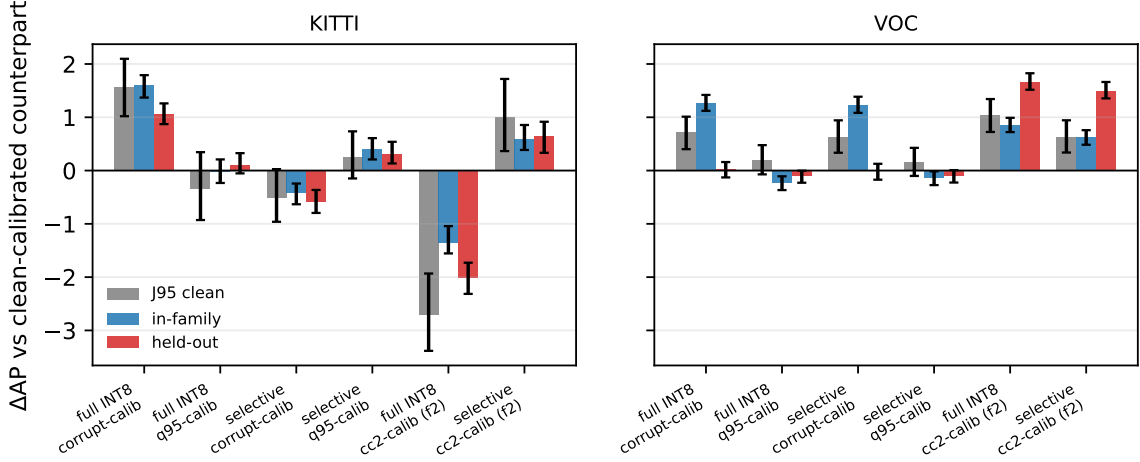
An earlier 144-cell exploratory grid analyzed the same paired YOLO11 engines—the same estimand ( $G_{\text{corr}} - G_{\text{J95}}$ ), codec control, pairing basis, and KITTI/VOC partitions—on four datasets including TT100K [66], differing in scope (the full COCO codec-control pool rather than the frozen 2,000-image subset, an extra dataset, and additional arms) and in the analysis plan (no Holm family, different summaries). It found a balanced additive interaction of  $-0.02$

AP  $[-0.24, +0.15]$  across the grid and  $-0.55 [-0.78, -0.29]$  on post-selection VOC/KITTI holdouts evaluated with independently selected checkpoints, with 56 point-estimate sign reversals after clean adjustment—17 with intervals excluding zero on the negative side. These findings motivated the frozen paired analysis reported above but are not pooled with it. Notably, the historical post-selection holdout estimate is the one piece of this evidence built on checkpoints selected outside the evaluation partition, and its negative sign is consistent with several of the paired blocks. Two exploratory-era transfer cases—RT-DETR-L [67, 68] and legacy RetinaNet—exhibited catastrophic INT8 clean failures under the unmigrated recipe, evidence in retrospect of recipe fragility rather than format fragility.

## 5. Discussion

### 5.1. Recipe and context, not the format label

The evidence layers converge on one conclusion: corruption robustness of a quantized detector is a property of the recipe, the model region, and the evaluation context, jointly with the format. No single layer could show this alone: the frozen paired layer shows the interaction has no stable sign (so pooling it into a format-level constant is meaningless); the diagnostic layer localizes *where* the level deficit lives (the regression head); and the intervention layer shows that the



**Figure 8:** Calibration variants versus their clean-calibrated counterparts (AP points, 95% paired-bootstrap intervals). Grey: J95 clean control; blue: in-family corruptions; red: held-out corruptions; “f2” groups swap the family roles (Fig. 2), and for the q95-calibration rows—which see no corrupted images—the in-family/held-out labels mark only the evaluation partition. Held-out bars that merely track the clean bar are not corruption-specific: that is the KITTI fold-1 pattern and, as a harm, the KITTI fold-2 full-INT8 pattern; the VOC fold-2 groups are the exception, where held-out clearly exceeds clean. Per-cell codec effects exist (a gain on the saturated KITTI motion-blur-s5 wall, a loss on a VOC noise cell, several gains on the KITTI selective tier) but stay under half an AP point in aggregate—a codec effect on shared failure cells, not evidence of corruption coverage.

*obvious remedy* (calibrating on corruptions) is itself context-dependent, transferring in one fold×dataset cell and harming another. Recipe choices—preprocessing contract, quantization sites—move clean AP by 21–42 AP points, several-fold larger than the largest corruption-induced  $\Delta E$  between FP8-matched and INT8-matched at the 12-cell corrupt-mean granularity (RetinaNet/KITTI: a  $-4.7$ -AP narrowing). Interactions between arms that additionally differ in recipe run higher still—up to  $+8.02$  AP on the in-family basis for the INT8-cc2calib–FP8-matched order on KITTI (a narrowing on an arm anchored about 20 AP below FP8, so scale dominates)—and at per-family granularity the FP8–INT8-matched interaction itself reaches  $-11.4$  AP (KITTI motion blur), so the “several-fold” comparison holds only at corrupt-mean granularity. Reporting “INT8” versus “FP8” robustness without the recipe contract is therefore not merely imprecise; it is capable of reversing conclusions.

## 5.2. Regression-head localization and possible mechanisms

The decomposition gives a clean localization (diagnostic-level, corroborated by the paired-protocol selective arm): classification-head quantization is harmless, regression-head quantization reproduces the collapse, and head exclusion restores clean-competitive accuracy. This is consistent with detector PTQ work emphasizing regression structures [27]. A plausible account, which our data are consistent with but do not isolate, is that box-regression outputs are unbounded deltas decoded through an exponential and a multiplication by the anchor size, so a fixed additive quantization error becomes a proportional localization error that AP at high IoU thresholds penalizes heavily, whereas classification scores pass through a

saturing sigmoid and are thresholded. The intervention adds the sharper negative: on KITTI, where the collapse lives, corruption-inclusive calibration does not repair the head and produces no corruption-specific held-out benefit in either fold—the second fold harms every level basis—so the failure is consistent with a region-level quantization-error sensitivity that none of the tested calibration distributions repairs—although our design varies only the calibration image distribution, so head-local range-management choices beyond entropy calibration remain untested. The small corruption-specific transfer observed on VOC fold 2 shows calibration coverage is a real but unstable contributor, which the context-dependent framing already predicts. FP8, which quantizes the same head with an exponent field and therefore a wide dynamic range, holds 49.19 AP (the pre-study FP8-legacy engine) and 49.48 AP (FP8-matched) on KITTI clean inputs, which also fits a dynamic-range account. The operand factorial sharpens this to the activation pathway: within the head, INT8 weights with fp16 activations recover 100%/103% of the matched-to-selective gain (KITTI/VOC) while fp16 weights over INT8 activations are statistically indistinguishable from the fully quantized arm ( $-0.04$  [ $-0.20, +0.11$ ] KITTI,  $+0.01$  [ $-0.08, +0.10$ ] VOC clean-level contrasts). INT8 storage of the head weights carries at most a residual penalty under  $\sim 0.2$  AP (detectable on KITTI only once activations are restored); the mechanism sits in the activation tensors, consistent with the unbounded-delta account above—though whether the operative INT8 cost there is tail clipping or bin resolution under the entropy-calibrated grid remains untested. The cross-architecture check is at least consistent with a head-shape account: FCOS’s regression head emits per-pixel distances decoded linearly through an

anchor-size scaling—without the exponential anchor-delta decode—and develops no head-specific INT8 deficit under the same recipe (Section 4.3, Table 7). That consistency is weak evidence, not confirmation: FCOS also differs in anchor density and centerness scoring, its selective arm dequantizes an extra output, and its head activations were never measured—and the exponential decode in the account above runs after the engine, downstream of the activations the factorial implicates, so the mechanism remains at the operand level. Why the head-exclusion effect is about 8× larger on KITTI than on VOC (21.3 versus 2.6 AP on the J95 basis) remains open; dataset-dependent target statistics (box-size distribution, class prior, average object scale) are the likely moderators.

### 5.3. Practical guidance

For deployment recipes, the evidence supports a concrete checklist: (i) verify the calibration preprocessing contract byte-for-byte against inference before interpreting any quantization gap; (ii) under matched contracts, evaluate the regression head and classification head as separate quantization regions, in the spirit of sensitivity-based mixed-precision assignment [69]—head exclusion recovered > 20 AP on KITTI RetinaNet while promoting only the `/head/regression_head/` subgraph to floating point, but bought nothing measurable on FCOS (selective minus matched J95 0.00 [−0.15, +0.12]/+0.01 [−0.04, +0.08]), so the split is worth evaluating, not assuming; (iii) do not rely on corruption-augmented calibration to transfer: across two folds its held-out effect ranged from a significant harm to a +1.66-AP gain depending on the dataset×family partition, and codec-match the calibration input encoding to the deployment stream before attributing gains to corruption coverage; (iv) report paired interactions together with absolute corrupted AP and the clean gap, so floor effects cannot masquerade as robustness. Selective precision is a retraining-free intervention worth evaluating first: it matched FP8-matched within intervals on KITTI and trailed it by +0.42 J95-clean / +0.96 corrupt AP on VOC. On deployment cost we make no latency or throughput claim: wall-clock latency measured on our shared GPU (graphics processing unit) host is inadmissible as evidence under the frozen protocol, and engine-build determinism was not audited. The repair is not free in principle: the promoted subgraph covers the regression-head nodes, which a layerwise MAC count of the torchvision graph (`analysis/regression_head_macs.py`) puts at 24.5–25.1% of convolution MACs at the frozen input size for the two reported class counts. All remaining quantization sites stay INT8-configured in the recipe; kernel-level execution rates were not measured. The W-only variant—head weights stay in INT8 storage, only head activations are promoted to fp16—is indistinguishable from full head exclusion on VOC and within 0.2 AP of it on KITTI, so a deployment that keeps head weights in INT8 storage loses nothing measurable beyond that bound (fp16 activation compute is still required, as under selective exclusion).

### 5.4. Limitations

*Artifact and engine scope.* The evidence is conditional on recorded artifacts: one checkpoint per dataset–family cell, one engine build per arm, one calibration method (entropy) and toolchain (ModelOpt/TensorRT) on the paired and intervention arms—the diagnostic sweeps additionally vary the estimator and the 128/512-image budget—and one runtime environment (TensorRT 11.1, RTX 5090); other PTQ methods may place quantization error differently. The mechanism analysis covers one architecture family deeply (RetinaNet, anchor-based) with a cross-family check on one anchor-free detector (FCOS), where the deficit does not occur; RT-DETR engines exist only under the legacy recipe and were excluded from the paired protocol, and the original head-region decomposition arms are diagnostic single points without intervals (the post-hoc operand-factorial and FCOS arms do carry paired intervals).

*Uncertainty coverage.* Bootstrap intervals cover image resampling only—not training-seed, calibration-draw, engine-build, or corruption-realization variance—and treat images as independent; KITTI frames drawn from shared driving sequences are correlated, so intervals there may be optimistic, and a sequence-clustered bootstrap is not computable from the retained manifests because drive metadata is not preserved in the evaluation ledger. Holm adjustment covers the nine paired YOLO11 blocks; RetinaNet, intervention, operand-factorial, and FCOS intervals are per contrast and unadjusted. Within the FCOS extension’s 56 contrast cells, 12 intervals exclude zero. Under a Holm correction across those cells ( $\alpha = 0.05/56 \approx 9 \times 10^{-4}$  for the smallest), the four deepest cells (bootstrap  $p < 0.001$ ; normal-approximation  $z \geq 3.5$ ) would survive—the KITTI FP8–INT8 corrupt mean among them—while the remaining eight ( $p = 0.002$ – $0.049$ , including the starred Table 7 cell and the two VOC FP8–INT8 per-family  $\Delta E$  cells cited in §4.3) would not; under a conservative plus-one bootstrap convention no FCOS cell survives. FCOS statements in the text are therefore unadjusted interval statements, and we flag them as such. Two-sided bootstrap  $p$ -values have resolution 1/1000. The null-seeking selective-versus-FP8 criterion bounds the residual but specifies no equivalence margin, so an interval containing zero is not a proof of equivalence [70].

*Provenance and protocol scope.* Three artifact-provenance caveats bound the paired layer: the KITTI and VOC evaluation partitions coincide with the checkpoint-selection partitions for all three detector families (and the COCO YOLO11 models use the official pretrained checkpoint), so absolute AP levels are optimistic; both paired arms share the selected checkpoint, but whether the two arms’ selection biases differ—and hence how much enters the gap and the interaction—is unquantified; and the YOLO11 engines predate the frozen contract and carry TensorRT’s default TF32 state, while the RetinaNet quantization arms were built with TF32 disabled

(the RetinaNet FP32 reference and legacy arms are TF32-on trtexec builds, but the  $\Delta E$  contrasts only join quantized arms built under the same policy), so the families also differ in builder policy (FCOS is internally TF32-off; Supplement). The paired protocol’s freeze covers the analysis contract (estimands, conditions, bootstrap schedule), not the artifacts it runs on—the nine-block Holm adjustment is an analysis-time choice implemented in the reporting code rather than a line in the frozen configuration. Historical exploratory results share the paired YOLO11 engines; only their retrained-holdout and legacy-transfer arms use different builds, and all are reported only as context.

*Corruption suite and intervention scope.* The corruption suite is synthetic and four types wide; severities order settings within a type and are not comparable across types, and the corrupt mean’s equal weighting includes the saturated motion-blur-s5 wall cell (reported alongside a leave-one-out sensitivity). The intervention uses two complementary folds over these types; other partitions and non-synthetic shifts are untested, and each calibration cell saw only 75–102 images. The intervention arms also differ on the clean control (the corrupt-calibration gains include a significant uniform clean-AP shift), which is why the corruption-specific reading relies on the held-out  $\Delta E$  decomposition rather than the raw held-out gain.

*Registration and anchoring caveats.* The fold-1 frozen criteria were a held-out significance test and a selective-versus-FP8 residual test—the magnitude comparison and held-out  $\Delta E$  analysis followed as interpretive—while fold 2 was frozen after that analysis and pre-specifies both rules, and the codec controls attached to the fold-1 contrasts are post-hoc additions. The fold partition also confounds corruption family with spectral character (fold 1 calibrates the high-frequency families and holds out the low-frequency ones), “held-out JPEG” is not fully unseen in codec terms because every calibration image is already JPEG quality-95 encoded, and the many intervention contrasts carry no familywise multiplicity correction, so individual significant cells should be read against the full table rather than in isolation. On the RetinaNet arms the J95 clean control departs from original clean by up to 0.8 AP (the fold-2 selective arm), larger than the  $\sim 0.3$ -point fidelity seen on YOLO11—enough to move small selective-tier residuals across significance boundaries on the original-clean basis (retained in the ledger). KITTI’s selective-arm numbers are development-contaminated; VOC serves as the cleaner check, and the two-dataset scope limits the localization’s generality.

*Evaluation scope.* COCO evidence is restricted to the frozen 2,000-image subset, and AP is a single utility summary; downstream risk thresholds belong to the application.

## 6. Conclusion

A paired, codec-controlled protocol with a common-image bootstrap turns the question “is INT8 or FP8 more robust to corruption” into a question with a defensible answer: on the corruption-induced change in the gap there is no stable winner—the interaction changes sign across datasets and scales, while FP8 exceeds INT8 on absolute clean and corrupted AP in every matched YOLO11 block mean (the FCOS blocks reverse that ordering on the corrupt means—significantly before adjustment on KITTI, marginally on VOC—while on the clean basis only the KITTI reversal excludes zero, marginally; all FCOS intervals are unadjusted), and recipe and model region determine the size and site of the accuracy collapse. The RetinaNet case makes the localization concrete. On KITTI a controlled diagnostic preprocessing rebuild recovers 21.2 AP points (128-image calibration, single point) and a regression-head exclusion recovers a further 21.3 (both on the J95 basis)—a magnitude that carries the disclosed development-contamination caveat, with its direction replicated on VOC; and a two-fold corruption-aware calibration intervention, read through the same paired estimand, is itself context-dependent—transferring a small corruption-specific benefit in one fold $\times$ dataset cell and significantly lowering accuracy in another—so calibration coverage is a secondary, unstable contributor rather than a reliable repair. A post-hoc operand factorial inside the head resolves the separation to operand level—activation quantization carries essentially the whole deficit while INT8 head weights cost at most  $\sim 0.2$  AP, and an anchor-free FCOS head shows no head-specific deficit under the same protocol (its INT8 matched and selective arms are statistically indistinguishable; the FP32-minus-selective gap is  $+0.07/+0.19$  AP on the clean basis and  $+0.12/+0.30$  AP on the corrupt mean—a generic-quantization gap, not a head-specific one)—but the operative failure mode (range clipping versus bin resolution) remains untested, and whether the RetinaNet finding generalizes to other anchor-based heads is unknown. Quantization robustness under corruption should be evaluated and reported per recipe—with paired inputs, clean-gap adjustment, and absolute corrupted AP—rather than per format label.

## Data and code availability

The Supplementary Information restates the full frozen contract: estimands, condition schedule, arm registry, inference/evaluation constants, corruption-aware calibration schedule, bootstrap schedule, evidence layers, and a reporting checklist. Manuscript tables and figures are generated deterministically from frozen ledgers and manifests. Engine registries, calibration manifests (including the corrupt-calibration schedule), input manifests, metric records, and paired-bootstrap draw caches are hash-bound and retained in the accompanying compact-evidence package, from which the analysis source regenerates all reported numbers. The package is deposited in the versioned research archive under concept DOI <https://doi.org/10.5281/zenodo.22031663>,

which currently resolves to the prior CVIU-era release lineage and mints the Neural Networks release on publication. Redistribution covers manifests, metrics, and analysis code; raw datasets, trained checkpoints, raw predictions, and built TensorRT engines are not redistributed.

## Declaration of competing interest

The authors declare no competing interests.

## CRediT authorship contribution statement

**Dinh Thuan Nguyen:** Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing—original draft. **Lam Phuong Nguyen:** Validation, Data curation, Writing—review and editing. **Vinh Huy Nguyen:** Methodology, Visualization, Writing—review and editing. **Sy Vu Quang:** Validation, Writing—review and editing. **Mohan Rajesh Elara:** Supervision, Writing—review and editing. **Anh Vu Le:** Supervision, Project administration, Writing—review and editing.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the authors used Cognition Devin for analysis-code development, experiment orchestration, and manuscript drafting support; OpenAI Codex for code refactoring, validation support, consistency checks, and language editing; OpenAI ChatGPT for literature-discovery support and manuscript critique; and Anthropic Claude Code for independent manuscript critique and submission-file review. These systems were not listed as authors or contributors and did not autonomously train detectors, build engines, generate predictions, fabricate or alter research data, or make final scientific decisions. The authors verified cited sources and numerical claims, reviewed every retained change, reran relevant validation scripts, and take full responsibility for the article.

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